
GARCH Modelling of Cryptocurrencies

Robin Derek Jan Modderaar

(2600112)

July 2020

Bachelor Thesis Econometrics and Operations Research

Thesis commission:

Dr. L. Hoogerheide

Dr. A. Borowska (co-reader)

Abstract

As cryptocurrencies are increasingly accepted by society as a means of payment and investment instrument, the goal of this paper is to find the best fitting model for the daily log returns on three of the current top ten most popular cryptocurrencies. Eleven GARCH-type models and five distributions are fitted to each cryptocurrency by using the Maximum Likelihood Estimation method. Conclusions are drawn by using multiple in- and out-of-sample tests on the best fitting models, forecasts based on them and the validity of their value at risk estimates.

Keywords: daily log returns; GARCH; maximum likelihood; logarithmic scoring; value at risk

Contents

1	Introduction	3
2	Data	3
3	Models	5
3.1	GARCH Models	6
3.2	Estimation of Value at Risk	7
4	Estimation Methods	8
4.1	Maximum Likelihood Estimation	8
5	Evaluation Methods	9
5.1	Jarque-Bera Test	9
5.2	Information Criteria	9
5.3	Logarithmic Scoring Rule	10
5.3.1	Diebold Mariano Test	10
5.4	Valua at Risk Tests	10
5.4.1	Unconditional Coverage Test	10
5.4.2	Independence Test	11
5.4.3	Conditional Coverage Test	11
6	Results	11
6.1	In-Sample Measures	12
6.2	Out-of-Sample Measures	12
6.2.1	Logarithmic Scoring Rule	12
6.2.2	Value at Risk Tests	12
6.2.3	Diebold-Mariano Test	13
7	Discussion	14
8	Conclusion	15
9	Mistakes in the article	15
10	Appendix	16

1 Introduction

Cryptocurrencies are digital assets designed to work as a medium of exchange. Bitcoin, first released in 2009, was the first cryptocurrency. Since the release of Bitcoin, over 6000 alternative cryptocurrencies have been created. As cryptocurrencies are increasingly accepted by society as a means of payment and investment instrument, the goal of this paper is to provide GARCH-type modeling of three of the most popular cryptocurrencies where our analysis is based on the article of Chu et al. (2017). The cryptocurrencies used are: Bitcoin, Litecoin and Ripple. Eleven different GARCH-type models were fit to the daily log returns of the exchange rates of each of these three cryptocurrencies. The method of Maximum Likelihood Estimation was used for fitting. The goodness of fit was measured in terms of multiple different criteria. Conclusions are drawn regarding best fitting models, forecasts based on them and their ability to estimate the value at risk. This paper is structured as follows. Section 2 describes the data used and some of their summary statistics. Section 3 describes the eleven GARCH-type models fitted, the five distributions used and the estimation of the value at risk. Sections 4 and 5 describe respectively the estimation method used and multiple in- and out-of-sample measures. The results of fitting the models are represented in Section 6. The discussion and some conclusions are stated in Sections 7 and 8.

2 Data

The data that we used in our analysis were the historical daily global price indices of the particular cryptocurrencies and were extracted from Yahoo Finance (yahoo.finance.com). We used the adjusted closing prices in US Dollars from 22 June 2014 until 01 December 2019. The summary statistics of the adjusted closing prices give immediate insight in the diversity of the data and are shown in Table 1.

Table 1: Summary statistics of the exchange rates of Bitcoin and Ripple from 22 June 2014 - 01 December 2019.

Statistic	Bitcoin	Litecoin	Ripple
N	1901	1901	1901
Minimum	178.103	1.157	0.004
Q1	410.815	3.649	0.007
Median	1272.830	14.342	0.036
Mean	3706.269	44.045	0.227
Q3	6635.892	62.029	0.321
Maximum	19497.400	358.336	3.378
Skewness	1.024	1.951	3.604
Kurtosis	3.300	7.487	23.934
SD	3964.740	57.237	0.341
Variance	15719161.470	3276.063	0.117
CV	1.070	1.300	1.506
IQR	6225.077	58.380	0.314
JB-test	339.232	2800.829	38827.367

The summary statistics are largest for Bitcoin, followed by Litecoin and Ripple. However, for our research, we are more interested in the daily log returns of the exchange rates of the respective cryptocurrencies. We calculated the daily log returns using the following formula:

$$y_t = \log(p_t) - \log(p_{t-1}),$$

where p_t is the adjusted closing price from a cryptocurrency at day t .

The summary statistics of the daily log returns are represented in Table 2. The log returns are positively skewed, except for Bitcoin. The log returns are moderately skewed, except the highly skewed log returns of Ripple. The log returns are heavy tailed with kurtosis greater than that of the normal distribution. These properties are also reflected in the histograms of the daily log returns of the cryptocurrencies in Figure 1.

Table 2: Summary statistics of the daily log returns on the exchange rates of Bitcoin, Litecoin and Ripple from 22 June 2014 - 01 December 2019.

Statistic	Bitcoin	Litecoin	Ripple
N	1900	1900	1900
Minimum	-0.238	-0.515	-0.616
Q1	-0.012	-0.019	-0.020
Median	0.002	0.000	-0.002
Mean	0.001	0.001	0.002
Q3	0.018	0.018	0.018
Maximum	0.225	0.511	1.027
Skewness	-0.287	0.708	2.948
Kurtosis	8.206	16.798	46.082
SD	0.039	0.057	0.067
Variance	0.001	0.003	0.004
CV	26.258	40.504	33.893
IQR	0.030	0.037	0.038
JB-test	2171.510	15229.861	149686.909

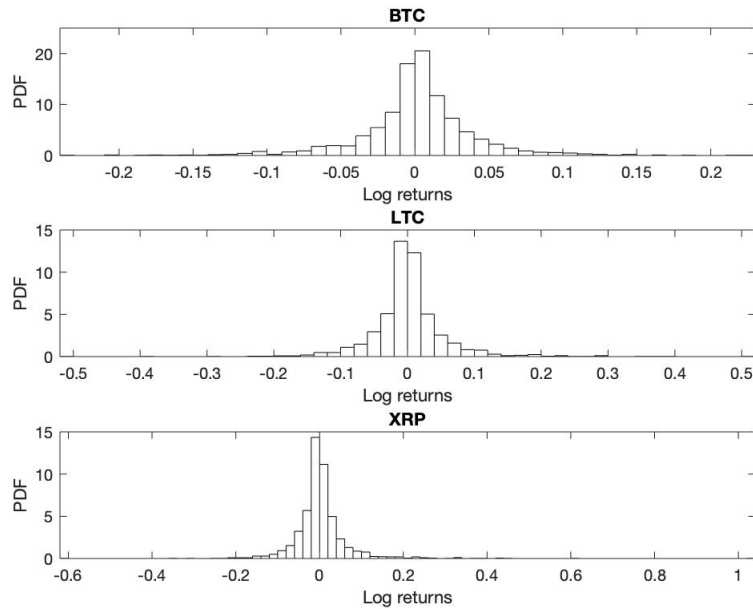


Figure 1: The histogram of the log returns on the exchange rates of Bitcoin, Litecoin and Ripple from 22 June 2014 - 01 December 2019.

The adjusted daily closing prices and the corresponding daily log returns of the three cryptocurrencies are displayed in Figure 2. The price index time series show clear signs of non-stationarity, as there is level change. There is an upward trend until December 2017, followed by the ‘2018 cryptocurrency crash’ in January 2018. The price of Bitcoin fell by about 65 percent during January 2018. Subsequently, nearly all other cryptocurrencies peaked from December 2017 until January 2018 and then followed the downward trend of Bitcoin. The log return time series appear to now be stationary, with obvious signs of meanreverting behaviour. However, cryptocurrencies are known to be highly volatile and this is also reflected in the plots of the log return time series.

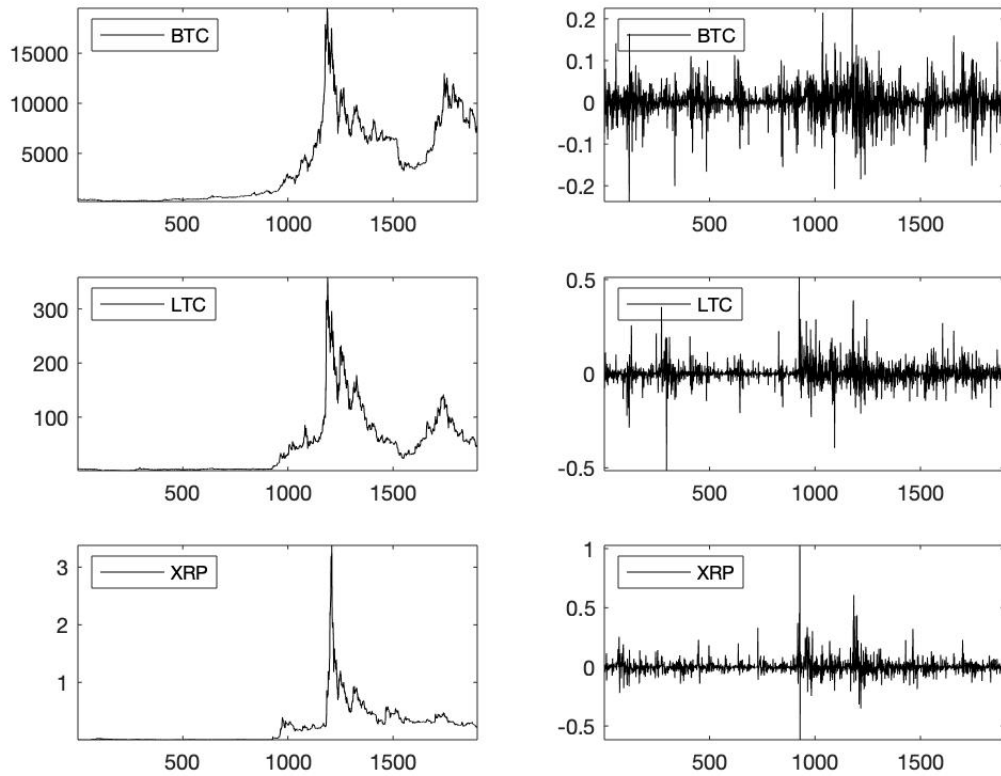


Figure 2: The plots of the daily exchange rates and the daily log returns on the exchange rates of Bitcoin, Litecoin and Ripple from 22 June 2014 - 01 December 2019.

3 Models

To analyse our cryptocurrency datasets, we used the following eleven GARCH type models: SGARCH(1,1), EGARCH(1,1), GJRGARCH(1,1), APARCH(1,1), IGARCH(1,1), CSGARCH(1,1), TGARCH(1,1), AVGARCH(1,1), NGARCH(1,1), NAGARCH(1,1) and ALLGARCH(1,1).

3.1 GARCH Models

Let X_t denote the daily log returns of respective cryptocurrency. Then, GARCH models can be specified as:

$$X_t = \mu_t + \sigma_t Z_t, \quad (1)$$

where μ_t is the conditional mean which we set at $\mu_t = 0$, and σ_t is a volatility process. Each GARCH model used in this paper uses the above specification in (1), but with a differing volatility process σ_t . For each GARCH-type model, X_t is allowed to follow one of the five distributions; Normal distribution, Skew Normal distribution (Azzalini (1985)), Student's t distribution (Gosset (1908)), Skew Student's t distribution (Fernandez and Steel (1998)) and Johnson's SU distribution (Johnson (1949)).

The standard GARCH model (Bollerslev (1986)), denoted by SGARCH(1,1) has:

$$\sigma_t^2 = \omega + \alpha_1 X_{t-1}^2 + \beta_1 \sigma_{t-1}^2$$

The integrated GARCH model (Engle and Bollerslev (1986)), denoted by IGARCH(1,1) is a particular case of the SGARCH(1,1) model for $\alpha_1 + \beta_1 = 1$.

The exponential GARCH model (Nelson (1991)), denoted by EGARCH(1,1), has:

$$\log(\sigma_t^2) = \omega + \alpha_1 Z_{t-1} + \gamma_1 |Z_{t-1}| + \beta_1 \log(\sigma_{t-1}^2)$$

The GJRARCH(1,1) model (Glosten et al. (1993)) has:

$$\sigma_t^2 = \omega + \alpha_1 X_{t-1}^2 + \gamma_1 I_{t-1} X_{t-1}^2 + \beta_1 \sigma_{t-1}^2,$$

where $I_{t-1} = 1$ if $X_{t-1} < 0$ and $I_{t-1} = 0$ else. The ALLGARCH(1,1) model (Hentschel (1995)) has:

$$\sigma_t^2 = (\omega + \alpha_1 (\sigma_{t-1}^2)^{\delta/2} [|Z_{t-1} - \eta_1| - \gamma_1 (Z_{t-1} - \eta_1)]^\delta + \beta_1 (\sigma_{t-1}^2)^{\delta/2})^{2/\delta}$$

The ALLGARCH(1,1) model contains the following as particular cases: NAGARCH(1,1) model (Engle and Ng (1993)) for $\delta = 2$ and $\gamma_1 = 0$; TGARCH(1,1) model (Zakoian (1994)) for $\delta = 1$ and $\eta_1 = 0$; NGARCH(1,1) model (Higgins and Bera. (1992)) for $\gamma_1 = \eta_1 = 0$.

The asymmetric power ARCH model (Ding et al. (1993)), denoted by APARCH(1,1), has:

$$\sigma_t^\delta = \omega + \alpha_1 (|X_{t-1}| - \gamma_1 X_{t-1})^\delta + \beta_1 \sigma_{t-1}^\delta$$

The APARCH(1,1) model contains the following as particular case: AVGARCH(1,1) model (Taylor (1986)) for $\delta = 1$ and $\gamma_1 = 0$.

The component standard GARCH model (Lee and Engle (1999)), denoted by CSGARCH(1,1), has:

$$\sigma_t^2 = q_t + \alpha_1 (X_{t-1}^2 - q_{t-1}) + \beta_1 (\sigma_{t-1}^2 - q_{t-1})$$

where:

$$q_t = \omega + \rho q_{t-1} + \phi (X_{t-1}^2 - \sigma_{t-1}^2)$$

All parameter specifications of the eleven GARCH models are displayed in Table 3.

Table 3: Parameter specifications GARCH models.

Model	ω	α_1	β_1	γ_1	δ	η_1	ϕ	ρ
SGARCH(1,1)	> 0	≥ 0	≥ 0	0	2	0	0	0
EGARCH(1,1)	$(-\infty, \infty)$	> 0	> 0	> 0	0	0	0	0
GJRGARCH(1,1)	> 0	> 0	> 0	> 0	0	0	0	0
APARCH(1,1)	> 0	≥ 0	≥ 0	$(-1, 1)$	> 0	0	0	0
IGARCH(1,1)	> 0	> 0	$1 - \alpha_1$	0	0	0	0	0
CSGARCH(1,1)	> 0	≥ 0	≥ 0	0	> 0	0	≥ 0	< 1
TGARCH(1,1)	> 0	≥ 0	≥ 0	$(-1, 1)$	1	0	0	0
AVGARCH(1,1)	> 0	≥ 0	≥ 0	0	1	0	0	0
NGARCH(1,1)	> 0	≥ 0	≥ 0	0	> 0	0	0	0
NAGARCH(1,1)	> 0	≥ 0	≥ 0	0	0	$(-\infty, \infty)$	0	0
ALLGARCH(1,1)	> 0	≥ 0	≥ 0	$(-1, 1)$	> 0	$(-\infty, \infty)$	0	0

3.2 Estimation of Value at Risk

The Value at Risk (VaR) is measured by assessing the amount of potential loss under a certain probability and over a certain period of time. Here, we provide formulas for estimating the one-day-ahead VaR for the five distributions considered in this paper.

For the normal distribution,

$$\widehat{VaR}_p = \hat{\mu}_{t+1} + \hat{\sigma}_{t+1} \Phi^{-1}(p),$$

where $\Phi(\cdot)$ denotes the cumulative distribution function of the standard normal distribution.

For the skew normal distribution (Azzalini (1985)),

$$\widehat{VaR}_p = \hat{\mu}_{t+1} + \hat{\sigma}_{t+1} F^{-1}(p),$$

where:

$$F(x) = \Phi(x) - 2T(x, \hat{\alpha})$$

for $-\infty < x < \infty$ and $-\infty < \alpha < \infty$, where $T(\cdot, \cdot)$ is Owens's T function (Owen (1956)).

For the Student's t distribution (Gosset (1908)),

$$\widehat{VaR}_p = \hat{\mu}_{t+1} + \hat{\sigma}_{t+1} F^{-1}(p),$$

where:

$$F(x) = \int_{-\infty}^x f(y) dy$$

for $-\infty < x < \infty$, where:

$$f(x) = \frac{\Gamma(\frac{v+1}{2})}{\Gamma(\frac{v}{2})} ((v-2)\pi)^{-1/2} \left(1 + \frac{x^2}{v-2}\right)^{-\frac{(v+1)}{2}}$$

for $-\infty < x < \infty$ and $v > 2$.

For the skew Student's distribution (Fernandez and Steel (1998)),

$$\widehat{VaR}_p = \hat{\mu}_{t+1} + \hat{\sigma}_{t+1} F^{-1}(p),$$

where:

$$F(x) = \int_{-\infty}^x f(y) dy$$

for $-\infty < x < \infty$, where:

$$f(x) = \frac{\Gamma(\frac{v+1}{2})}{\Gamma(\frac{v}{2})} ((v-2)\pi)^{-1/2} \times s \times \left(\frac{2}{\xi + \frac{1}{\xi}} \right) \times \left(1 + \frac{(s \times x + m)^2}{v-2} \xi^{-2I_t} \right)^{\frac{-(v+1)}{2}}$$

with:

$$I_t = \begin{cases} 1 & \text{if } s \times x + m \geq 0 \\ -1 & \text{if } s \times x + m < 0 \end{cases}$$

$$m = \frac{\Gamma(\frac{v-1}{2})}{\Gamma(\frac{v}{2})} \times \sqrt{\frac{v-2}{\pi}} \times \left(\xi - \frac{1}{\xi} \right)$$

$$s = \sqrt{\xi^2 + \left(\frac{1}{\xi^2} - 1 \right) - m^2}$$

for $-\infty < x < \infty$, $v > 2$ and $\xi > 0$.

For the Johnson's SU distribution (Johnson (1949)),

$$\widehat{VaR}_p = \hat{\mu}_{t+1} + \hat{\sigma}_{t+1} F^{-1}(p),$$

where:

$$F(x) = \Phi\left(\hat{\alpha} + \hat{\beta} \operatorname{arcsinh}(x)\right)$$

for $-\infty < x < \infty$, $\alpha > 0$ and $\beta > 0$, with:

$$f(x) = \frac{\delta}{\lambda \sqrt{1 + \left(\frac{x-\xi}{\lambda}\right)^2}} \phi\left[\gamma + \delta \sinh^{-1}\left(\frac{x-\xi}{\lambda}\right)\right],$$

for $-\infty < x < \infty$, $\xi > 0$, $\lambda > 0$, $\gamma > 0$ and $\delta > 0$.

4 Estimation Methods

The parameters of the GARCH(1,1) models and the five distributions introduced in previous sections were estimated by the method of Maximum Likelihood Estimation (MLE).

4.1 Maximum Likelihood Estimation

The parameters of the GARCH models and their innovations used in this paper were estimated by maximizing the log likelihood function divided by the number of observations T , defined as:

$$l(x_1, \dots, x_T | \theta) = \frac{1}{T} \sum_{t=1}^T \log f(x_t | \theta, x_{t-1}, x_{t-2}, \dots),$$

where x_t are the daily log-returns on the respective cryptocurrency, θ is the set of parameters and $f(x)$ is the pdf of one of the five distributions.

5 Evaluation Methods

5.1 Jarque-Bera Test

To check the daily log returns on the three cryptocurrencies for normality, we used the Jarque-Bera (JB) test (Jarque and Bera (1980)). The JB-test is a goodness-of-fit test of whether the dataset has the kurtosis and skewness matching a normal distribution. The JB-test statistic defined is as:

$$JB = \frac{N}{6} \left(Skewness^2 + \frac{1}{4} (Kurtosis - 3)^2 \right),$$

where N is the number of observations. Normality is rejected if the test statistic exceeds the critical value, which is at a 5% significance level and for $N = 1900$, equal to 5.9619 (MATLAB jbtest).

5.2 Information Criteria

All of the GARCH(1,1) models used in this paper were fitted by the maximum likelihood method. Discrimination among the models was performed using various criteria:

- the Akaike information criterion by Akaike (1974) defined as:

$$AIC = 2k - 2 \log L(\hat{\Theta}),$$

where k is the number of unknown parameters, Θ the vector of the unknown parameters, $\hat{\Theta}$ their maximum likelihood estimates and $\log L(\cdot)$ is the log likelihood function;

- the Bayesian information criterion by Schwarz (1978) defined as:

$$BIC = k \log n - 2 \log L(\hat{\Theta}),$$

where n is the number of observations;

- the Consistent Akaike Information Criterion (CAIC) by Bozdogan (1987) defined as:

$$CAIC = -2 \log L(\hat{\Theta}) + k(\log n + 1);$$

- the corrected Akaike Information Criterion (AICc) by Hurvich and Tsai (1989) defined as:

$$AICc = AIC + \frac{2k(k+1)}{n-k-1};$$

- the Hannan-Quinn criterion by Hannan and Quinn (1979) defined as:

$$HQC = -2 \log L(\hat{\Theta}) + 2k \log \log n.$$

Model selection is based on the lowest values for these criteria. The lower the values, the better the fit.

5.3 Logarithmic Scoring Rule

The Logarithmic Scoring Rule (LSR) is used to compare the predictive qualities of the eleven GARCH models, whereby the different distributions are taken into account as well. The LSR (Diks et al. (2011)) can be seen as an out-of-sample log likelihood. The parameters are estimated with the in-sample data, subsequently the log likelihood is evaluated using the out-of-sample data. The LSR is computed by taking the average of all the log density forecasts $\hat{f}_t$. The model combined with one of the five distributions with the highest LSR predicts the best.

Since the out-of-sample estimation of the log density gets worse when the model is overfitted, there is no need for a ‘penalty term’ for the number of parameters used in the LSR. It’s possible for a ‘smaller’ model to have a better out-of-sample estimated log density, than a ‘bigger’ model, as long as the restrictions for the smaller model are correct. The in-sample criteria in section 5.2 do have a penalty term, as the in-sample quality doesn’t get worse due to overfitting, even when the added parameters in the ‘bigger’ model are useless.

5.3.1 Diebold Mariano Test

The Diebold Mariano (DM) test (Diebold and Mariano (1995)) is used to check whether model 1 is significantly better (or worse) than model 2. The DM-test is calculated by first taking the difference between the log density forecasts of the two models:

$$d_{12,t} = \hat{f}_t^1 - \hat{f}_t^2,$$

then,

$$DM_{12} = \frac{\bar{d}_{12}}{\hat{\sigma}_{\bar{d}_{12}}}$$

where $\bar{d}_{12} = \frac{1}{T} \sum_{t=1}^T d_{12,t}$ and $\hat{\sigma}_{\bar{d}_{12}}$ is a consistent estimate of the standard deviation of $\bar{d}_{12}$.

Under the null hypothesis $H_0 = \mathbb{E}[\bar{d}_{12}] = 0$, DM_{12} is rejected at a 5% significance level if its value lies outside $(-1, 960; 1, 960)$.

5.4 Value at Risk Tests

We applied three different backtests to determine the accuracy of the in section 3.2 defined estimates of the one-day-ahead Value at Risk (VaR). Backtesting is a technique that compares the estimated VaR with the actual losses in the out-of-sample data.

5.4.1 Unconditional Coverage Test

For the LR test of Unconditional Coverage (Christoffersen (1998)), we first have to obtain the indicator sequence, $\{I_t\}_{t=1}^T$:

$$I_t = \begin{cases} 1 & \text{if } x_t < VaR_t(p) \\ 0 & \text{if } x_t \geq VaR_t(p) \end{cases},$$

where T is the total number of out-of-sample values for which we estimate the VaR and p is the probability upon which we estimated the VaR.

Second, $H = \sum_{t=1}^T I_t$ is computed, where H is the number of days the out-of-sample losses

exceed the estimated VaR.

Then the Likelihood Ratio statistic for UC, under $H_0 : p = \alpha$, is defined as:

$$LR_{UC} = -2 \log[(1 - \alpha)^{T-H} \alpha^H] + 2 \log[(1 - \frac{H}{T})^{T-H} (\frac{H}{T})^H] \sim \chi^2(1)$$

LR_{UC} values above the critical value should lead to rejection of H_0 and indicate an inaccurate model.

5.4.2 Independence Test

The Unconditional Coverage Test is one part of the Conditional Coverage Test. The other part is the Independence Test (Christoffersen (1998)).

Where the Unconditional Coverage Test only tests for the fraction of exceedances. The Independence Test, tests for clustering of the exceedances. To do this, I_t can be represented as a Markov chain with two states:

$$\Pi = \begin{bmatrix} 1 - \pi_{01} & \pi_{01} \\ 1 - \pi_{11} & \pi_{11} \end{bmatrix}$$

where π_{ij} is the probability of i at time $t - 1$ being followed by j at time t . Further, n_{ij} is defined as the number of observations in $\{I_t\}_{t=1}^T$, in which j succeeded i . We estimate Π as:

$$\hat{\Pi} = \begin{bmatrix} \frac{n_{00}}{n_{00}+n_{01}} & \frac{n_{01}}{n_{00}+n_{01}} \\ \frac{n_{10}}{n_{10}+n_{11}} & \frac{n_{11}}{n_{10}+n_{11}} \end{bmatrix}$$

The null hypothesis is defined as: $H_0 : \pi_{01} = \pi_{11}$, with the test statistic:

$$LR_{IND} = -2 \log[(1 - \frac{H}{T})^{T-H} (\frac{H}{T})^H] + 2 \log[(1 - \hat{\pi}_{01})^{n_{00}} \hat{\pi}_{01}^{n_{01}} (1 - \hat{\pi}_{11})^{n_{10}} \hat{\pi}_{11}^{n_{11}}] \sim \chi^2(1),$$

5.4.3 Conditional Coverage Test

As stated above, the Conditional Coverage Test is a combination of the Unconditional Coverage Test and the Independence Test (Christoffersen (1998)). The Conditional Coverage Test accounts both for the fraction of exceedances and clustering.

The test statistic is defined as:

$$LR_{CC} = LR_{UC} + LR_{IND} \sim \chi^2(2).$$

A rejection of H_0 is an indication of too many exceedances and/or clustering of the exceedances.

6 Results

The SGARCH(1,1), EGARCH(1,1), GJRGARCH(1,1), APARCH(1,1), IGARCH(1,1), CSGARCH(1,1), TGARCH(1,1), AVGARCH(1,1), NGARCH(1,1), NAGARCH(1,1) and the ALLGARCH(1,1) models were fitted to the daily log returns of Bitcoin, Litecoin and Ripple. The distribution of the innovations process were taken to be one of Normal distribution, Skew Normal distribution, Student's t distribution, Skew Student's t distribution or Johnson's SU distribution.

The results for the Maximum Likelihood are shown in Tables 6-8 in the Appendix. The Jarque-Bera test statistics are shown in Table 2. For all three cryptocurrencies, the critical value is well

exceeded, so we reject the null hypothesis for being normally distributed. That the Normal distribution probably isn't the optimal distribution could also be concluded by looking at the skewness and kurtosis in Table 2. The initial 1900 daily log returns were split in an in-sample data set (first 1500) and an out-of-sample data set (last 400).

6.1 In-Sample Measures

The values of Loglikelihood, AIC, BIC, CAIC, AICc and HQC are given in Tables 9-11 in the Appendix for the fitted models. For all cryptocurrencies the Skew Normal distribution and the Johnson's SU distribution give the lowest values for AIC, BIC, CAIC, AICc, HQC, and the highest values for Loglikelihood.

For Bitcoin, the Skew Normal NAGARCH and the Johnson's SU EGARCH appear to give the best fit. For Litecoin, the Skew Normal GJRGARCH and the Johnson's SU EGARCH. For Ripple, the Skew Normal GJRGARCH, the Skew Normal NAGARCH and the Johnson's SU EGARCH.

6.2 Out-of-Sample Measures

6.2.1 Logarithmic Scoring Rule

The Logarithmic Scoring Rule results are given in Tables 9-11. Generally, the Skew Normal and the Johnson's SU distributions appear to give the highest values for the LSR. The Skew Normal SGARCH and the Skew Normal NAGARCH result in the highest LSR values for Bitcoin. The Johnson's SU EGARCH, Skew Normal EGARCH and Skew Normal GJRGARCH for Litecoin. And, the Skew Normal GJRGARCH, Johnson's SU EGARCH and Skew Normal IGARCH for Ripple.

6.2.2 Value at Risk Tests

For all three cryptocurrencies, the Unconditional Coverage (UC), Independence (IND) and Conditional Coverage (CC) test statistics were computed at 0.05 and 0.01 levels of significance. The test statistics and corresponding p-values are given in Tables 17-19.

Regarding Bitcoin, all p-values of the models following the Normal distribution are significant at both 0.05 and 0.01, except the EGARCH and TGARCH models. Further, only the Skew Student's t IGARCH and the Johnson's SU SGARCH models appear to be significant at both levels.

Regarding Litecoin, all p-values of the models following the Normal distribution are significant at both 0.05 and 0.01, except the AVGARCH and NAGARCH models. Further, only the Skew Normal TGARCH, Skew Normal NGARCH, Skew Student's t IGARCH models appear to be significant at both levels. The Skew Normal NAGARCH model is significant at 0.05, but not significant at 0.01.

Regarding Ripple, all p-values of the models following the Normal distribution are significant at both 0.05 and 0.01, except the EGARCH and TGARCH models. Further, only the Skew Normal NGARCH, Johnson's SU SGARCH, and Johnson's SU GJRGARCH models appear to be significant at both levels.

6.2.3 Diebold-Mariano Test

The Diebold-Mariano Test was computed to compare the different models per distribution and to compare different distributions per model. The results are given in Tables 12-16 and were used to see if the models that provide acceptable estimates of the VaR, see section 6.2.1, are significantly better than other models.

For Bitcoin, none of the Normally distributed models is significantly better or worse than any of the other Normally distributed models. The Skew Student's t IGARCH model only performs significantly better than the Skew Student's t CSGARCH model. Comparing the IGARCH model among different distributions, the Skew Normal IGARCH and the Johnson's SU IGARCH perform significantly better. The Johnson's SU SGARCH only performs significantly better than the Johnson's SU IGARCH model. Comparing the SGARCH model among different distributions, the Skew Normal SGARCH model performs significantly better.

For Litecoin, only the performance of the IGARCH model of the Normally distributed models is significantly worse than other models. The Skew Normal TGARCH model only performs significantly better than the Skew Normal NGARCH model. Comparing the TGARCH model among different distributions, the Skew Normal TGARCH model performs significantly better. The Skew Normal ALLGARCH model doesn't perform significantly better than any other model. Comparing the ALLGARCH model among different distributions, the Skew Normal ALLGARCH model performs significantly better. The Skew Student's t IGARCH model doesn't perform significantly better than any other model. Comparing the IGARCH model among different distributions, the Johnson's SU IGARCH and Skew Normal IGARCH perform significantly better.

For Ripple, for the Normally distributed models, only the APARCH model performs significantly better than the AVGARCH model. The SGARCH model performs significantly better than the IGARCH model. All other combinations are indifferent. The Johnson's SU GJRGARCH performs significantly better than the SGARCH, APARCH, IGARCH, NGARCH and ALLGARCH. Comparing the GJRGARCH among different distributions, the Skew Normal GJRGARCH performs significantly better.

For all three cryptocurrencies, even though the Normally distributed models provide acceptable estimates of the VaR, according to Tables 17-19 the other distributions perform significantly better.

Furthermore, we used the DM-Test results to compare the models that performed best according to the in-sample criteria and the Logarithmic Scoring Rule.

For Bitcoin, the Skew Normal SGARCH, the Skew Normal NAGARCH and the Johnson's SU EGARCH came out on top. The Skew Normal SGARCH and the Skew Normal NAGARCH perform indifferently. It is not possible to compare the models with the Johnson's SU EGARCH as the models are not indifferent regarding the distributions. For Litecoin, those are the Skew Normal GJRGARCH and the Johnson's SU EGARCH, of which the EGARCH performs best. For Ripple, again the Skew Normal GJRGARCH and the Johnson's SU EGARCH. However, they can not be compared as the models are not indifferent regarding the distributions.

7 Discussion

The results showed us that the Skewed Normal and the Johnson's SU distributions provided overall the best fits for the data. Specifically, for Bitcoin, the Skew Normal NAGARCH, the Skew Normal SGARCH and the Johnson's SU EGARCH. For Litecoin, the Johnson's SU EGARCH. For Ripple the Skew Normal GJRGARCH or the Johnson's SU EGARCH.

However, regarding the validity of the value at risk, the results turned out to be different. For each cryptocurrency, only a few models appeared to actually provide accurate values for the VaR and, against expectations, the Normal distribution outperformed the other distributions. Though, this could be explained by looking at the summary statistics of the in- and out-of-sample datasets in Tables 4 and 5. The in-sample period used for fitting the models, appears to be much wilder than the out-of-sample period: the sample variance and the sample kurtosis of the log returns are much higher in the in-sample period than in the out-of-sample period. Causing overestimation of the risk. The estimations of the variance and the VaR in the out-of-sample are now too big, because they are based on the 'wilder' in-sample period. The Normal distribution actually underestimates the risk. However, because the risk in the out-of-sample period is lower than in the in-sample period, the Normal distribution now accidentally performs best regarding forecasting.

Table 4: Summary statistics of the in- and out-of-sample exchange rates of Bitcoin, Litecoin and Ripple from 22 June 2014 - 01 December 2019.

Statistic	BTC in-sample	BTC out-of-sample	LTC in-sample	LTC out-of-sample	RIPP in-sample	RIPP out-of-sample
N	1500	400	1500	400	1500	400
Minimum	178.1030	3236.7617	1.1570	23.4643	0.0041	0.2186
Q1	361.1175	4007.7483	3.4148	46.0217	0.0065	0.2944
Median	666.1040	7175.4526	3.9982	59.8468	0.0088	0.3149
Mean	2837.5739	6954.2163	38.0975	66.3411	0.1983	0.3333
Q3	5626.3599	9351.7036	54.6448	84.5024	0.2543	0.3690
Maximum	19497.4190	13016.2314	358.3360	141.9030	3.3778	0.5491
Skewness	1.6420	0.1760	2.2030	0.7072	3.5816	0.9254
Kurtosis	5.2744	1.6305	7.8641	2.8468	21.2576	3.3621
SD	3784.2835	2757.0773	61.3726	28.6048	0.3778	0.0648
Variance	1.4321e+07	7.6015e+06	3766.5957	818.2355	0.1427	0.0042
CV	1.3336	0.3965	1.6109	0.4312	1.9052	0.1945
IQR	5265.2428	5343.9553	51.2300	38.4807	0.2478	0.0746
JB-test	997.3188	33.3233	2691.9992	33.7366	24040.7761	59.2705

Table 5: Summary statistics of the in- and out-of-sample daily log returns on the exchange rates of Bitcoin, Litecoin and Ripple from 22 June 2014 - 01 December 2019.

Statistic	BTC in-sample	BTC out-of-sample	LTC in-sample	LTC out-of-sample	RIPP in-sample	RIPP out-of-sample
N	1499	399	1499	399	1499	399
Minimum	-0.2376	-0.1518	-0.5146	-0.1803	-0.6163	-0.1342
Q1	-0.0119	-0.0143	-0.0169	-0.0268	-0.0207	-0.0185
Median	0.0023	0.0011	-0.0006	-0.0012	-0.0030	-0.0023
Mean	0.0018	0.0004	0.0016	-0.0001	0.0030	-0.0017
Q3	0.0179	0.0158	0.0164	0.0247	0.0193	0.0148
Maximum	0.2251	0.1600	0.5114	0.2687	1.0274	0.2286
Skewness	-0.3574	-0.0115	0.7264	0.5334	2.8875	0.9254
Kurtosis	8.6248	6.4615	17.9524	6.6416	42.0785	7.0095
SD	0.0390	0.0380	0.0589	0.0503	0.0722	0.0394
Variance	0.0015	0.0014	0.0035	0.0025	0.0052	0.0016
CV	22.0657	83.6890	37.8122	-314.7036	24.3759	-22.7486
IQR	0.0298	0.0301	0.0333	0.0515	0.0400	0.0333
JB-test	2007.9778	199.2090	14095.8906	239.3849	97464.9309	284.8909

8 Conclusion

Summarizing, eleven different GARCH-type models and five different distributions were fitted to the daily log returns of Bitcoin, Litecoin and Ripple. We find that the Skewed Normal and the Johnson's SU distributions in general provide the best fit. However, regarding the validity of the value at risk, the Normal distribution came out on top. Furthermore, the Skew Normal NAGARCH, Skew Normal SGARCH or the Johnson's SU EGARCH provided the best fit for Bitcoin. The Johnson's SU EGARCH for Litecoin. And, the Skew Normal GJRGARCH or Johnson's SU EGARCH for Ripple. Suggestions for future work are to include more cryptocurrencies and to include more types of models as only a limited number of models is used in this paper. For example, multivariate GARCH-type models to describe the joint behaviour of the cryptocurrencies. Besides using the Maximum Likelihood Estimation method for parameter estimation, one could try Bayesian Estimation. Another possibility is to use the expected shortfall in addition to the value at risk.

9 Mistakes in the article

The following mistakes were made in the article of Chu et al. (2017):

- SGARCH(1,1): Z_{t-1}^2 should be $(X_{t-1} - \mu_{t-1})^2$.
- GJR-GARCH(1,1): Z_{t-1}^2 should be $(X_{t-1} - \mu_{t-1})^2$.
- APARCH(1,1): $|Z_{t-1}|$ and Z_{t-1} should be $|X_{t-1} - \mu_{t-1}|$ and $(X_{t-1} - \mu_{t-1})$.
- CSGARCH(1,1): Z_{t-1}^2 should be $(X_{t-1} - \mu_{t-1})^2$.
- The pdfs of the Student t and skew Student t distributions in the article should be replaced with the pdfs described in section 3.2.

10 Appendix

Table 6: Maximum Likelihood parameter estimates for Bitcoin

Model	ω	α_1	β_1	γ_1	δ	η_1	ϕ	ρ	Dist.
Norm. SGARCH	0.0001	0.1738	0.7742						
Norm. EGARCH	-0.4224	0.0000	0.9611	0.2300					
Norm. GJRGARCH	0.0002	0.2800	0.5920	0.0000					
Norm. APARCH	0.0033	0.1312	0.8711	0.0724	0.7346				
Norm. IGARCH	0.0001	0.2540	0.7460						
Norm. CSGARCH	0.4995	0.1500	0.8500				0.4998	0.4999	
Norm. TGARCH	0.0026	0.1293	0.8752	0.7876					
Norm. AVGARCH	0.0012	0.1354	0.8746						
Norm. NGARCH	0.0383	0.1905	0.7168		0.3854				
Norm. NAGARCH	0.0001	0.1762	0.7730			0.0810			
Norm. ALLGARCH	0.0148	0.1306	0.8450	0.1590	0.3999	-0.0325			
Skew. Norm. SGARCH	0.0010	0.9518	0.0005						$\lambda = -6.6548$
Skew. Norm. EGARCH	-1.9393	0.5491	0.7820	0.5020					$\lambda = 3.0620$
Skew. Norm. GJRGARCH	0.0100	0.0007	0.0000	0.0003					$\lambda = 16.2049$
Skew. Norm. APARCH	0.0108	0.3003	0.2473	0.1222	0.8207				$\lambda = -2.1958$
Skew. Norm. IGARCH	0.0002	0.7803	0.2197						$\lambda = 3.0702$
Skew. Norm. CSGARCH	0.0149	0.5837	0.4100				0.3965	0.1578	$\lambda = 3.0551$
Skew. Norm. TGARCH	0.0078	0.2365	0.2938	-0.1278					$\lambda = -3.7079$
Skew. Norm. AVGARCH	0.0152	0.0729	0.0772						$\lambda = -3.2000$
Skew. Norm. NGARCH	0.0003	0.2232	0.6913		0.7011				$\lambda = -2.0159$
Skew. Norm. NAGARCH	0.0044	0.0000	0.0003			1.9313			$\lambda = -11.3837$
Skew. Norm. ALLGARCH	0.0165	0.2647	0.6566	0.6314	0.7683	0.0832			$\lambda = -6.5379$
Stud. SGARCH	0.0001	1.0000	0.8144						$v = 2.1470$
Stud. EGARCH	-0.1607	0.0704	0.9954	0.2109					$v = 2.7336$
Stud. GJRGARCH	0.0001	1.0000	0.8152	0.0111					$v = 2.1589$
Stud. APARCH	0.0009	0.1652	0.8877	-0.1576	0.7067				$v = 2.7645$
Stud. IGARCH	0.0001	0.2580	0.7420						$v = 2.9597$
Stud. CSGARCH	0.4999	0.1502	0.8498				0.4993	0.4979	$v = 3.4894$
Stud. TGARCH	0.0003	0.1928	0.8800	-0.1219					$v = 2.7680$
Stud. AVGARCH	0.0004	0.2065	0.8714						$v = 2.7539$
Stud. NGARCH	0.0010	0.1850	0.8763		0.7674				$v = 2.7498$
Stud. NAGARCH	0.0001	1.0000	0.8143			-0.0538			$v = 2.1413$
Stud. ALLGARCH	0.0004	0.5218	0.8801	-0.0636	0.8190	-0.0275			$v = 2.0411$
Skew. Stud. SGARCH	0.0001	0.1313	0.5257						$v = 3.3448, \xi = 0.0001$
Skew. Stud. EGARCH	-0.2247	0.0307	0.9896	0.1065					$v = 3.9763, \xi = 0.0003$
Skew. Stud. GJRGARCH	0.0001	0.2443	0.3591	0.0000					$v = 3.1808, \xi = 0.0001$
Skew. Stud. APARCH	0.0002	0.0894	0.8819	-0.1852	0.9334				$v = 3.9739, \xi = 0.0013$
Skew. Stud. IGARCH	0.0001	0.3193	0.6807						$v = 2.3211, \xi = 0.0001$
Skew. Stud. CSGARCH	63.3524	0.4258	0.5742				0.5002	0.4971	$v = 122.6623, \xi = 0.4675$
Skew. Stud. TGARCH	0.0002	0.0872	0.8808	-0.1792					$v = 3.9768, \xi = 0.0002$
Skew. Stud. AVGARCH	0.0002	0.0916	0.8733						$v = 3.9839, \xi = 0.0002$
Skew. Stud. NGARCH	0.0006	0.8088	0.8474		1.2054				$v = 2.0305, \xi = 0.0155$
Skew. Stud. NAGARCH	0.0001	0.1339	0.5187			-0.2048			$v = 3.3011, \xi = 0.0003$
Skew. Stud. ALLGARCH	0.0043	0.1192	0.8514	-0.0271	0.4603	-0.2827			$v = 4.1721, \xi = 0.0634$
SU. SGARCH	3.1886	0.5415	0.6958						$\gamma = -0.8129, \delta = 8.0942, \xi = -0.0339, \lambda = 0.0358$
SU. EGARCH	22.6128	0.4992	0.4964	0.5010					$\gamma = 999.9680, \delta = 94.8459, \xi = 0.0000, \lambda = 0.0001$
SU. GJRGARCH	2.8439	0.5319	0.9901	0.5316					$\gamma = -0.1804, \delta = 5.9475, \xi = -0.0056, \lambda = 0.0061$
SU. APARCH	1.6362	0.5035	0.9531	0.0125	0.6562				$\gamma = 4.4291, \delta = 7.0232, \xi = -0.0001, \lambda = 0.0085$
SU. IGARCH	2.0581	0.4880	0.5120						$\gamma = -2.6103, \delta = 10.5775, \xi = -0.0827, \lambda = 0.1235$
SU. CSGARCH	40.8476	0.3035	0.0000				0.0006	0.9998	$\gamma = 45.0196, \delta = 8.6837, \xi = 0.0000, \lambda = 0.0003$
SU. TGARCH	9.2257	0.6379	0.9885	-0.0714					$\gamma = 9.4407, \delta = 4.5916, \xi = 0.0000, \lambda = 0.0001$
SU. AVGARCH	40.5695	0.5637	0.9950						$\gamma = 39.9095, \delta = 7.4095, \xi = 0.0000, \lambda = 0.0001$
SU. NGARCH	1.7828	0.4555	0.8279		0.7133				$\gamma = 4.4879, \delta = 7.0436, \xi = -0.0004, \lambda = 0.0086$
SU. NAGARCH	2.4371	0.5329	0.6977			0.8290			$\gamma = -1.0589, \delta = 8.0388, \xi = -0.0171, \lambda = 0.0361$
SU. ALLGARCH	1.6893	0.5497	0.6175	0.1150	0.6175	1.1603			$\gamma = 4.0957, \delta = 6.4287, \xi = -0.0012, \lambda = 0.0051$

Table 7: Maximum Likelihood parameter estimates for Litecoin

Model	ω	α_1	β_1	γ_1	δ	η_1	ϕ	ρ	Dist.
Norm. SGARCH	0.0001	0.0819	0.8774						
Norm. EGARCH	-0.4182	0.0758	0.9456	0.1548					
Norm. GJRGARCH	0.0008	0.4070	0.4901	0.0000					
Norm. APARCH	0.0139	0.0853	0.8918	-0.7963	0.4717				
Norm. IGARCH	0.0001	0.1225	0.8775						
Norm. CSGARCH	0.4918	0.1268	0.8731				0.4989	0.5075	
Norm. TGARCH	0.0032	0.1015	0.8787	1.0000					
Norm. AVGARCH	0.0032	0.1015	0.8787						
Norm. NGARCH	0.0010	0.0999	0.8774		1.3781				
Norm. NAGARCH	0.0001	0.0742	0.8336			-0.8820			
Norm. ALLGARCH	0.0429	0.2200	0.6304	-0.2339	0.5641	-0.1346			
Skew. Norm. SGARCH	0.0001	0.3545	0.5529						$\lambda = 3.1455$
Skew. Norm. EGARCH	-1.9421	0.5418	0.7807	0.4872					$\lambda = 3.0615$
Skew. Norm. GJRGARCH	0.0005	0.1767	0.0547	0.1786					$\lambda = 3.1078$
Skew. Norm. APARCH	0.1960	0.5359	0.0056	-0.0389	0.6018				$\lambda = -6.3410$
Skew. Norm. IGARCH	0.0002	0.8300	0.1700						$\lambda = 3.1320$
Skew. Norm. CSGARCH	0.0160	0.5841	0.4089				0.4101	0.1515	$\lambda = 3.0987$
Skew. Norm. TGARCH	0.0162	0.3071	0.2286	-0.0917					$\lambda = -1.8816$
Skew. Norm. AVGARCH	0.0273	0.0816	0.0877						$\lambda = 3.2561$
Skew. Norm. NGARCH	0.0350	0.5378	0.4024		0.6219				$\lambda = -2.0734$
Skew. Norm. NAGARCH	0.0022	0.9686	0.0005			0.9158			$\lambda = -6.3112$
Skew. Norm. ALLGARCH	0.0264	0.1667	0.2225	-0.1163	0.8992	1.0283			$\lambda = -2.0594$
Stud. SGARCH	0.0001	1.0000	0.8545						$v = 2.0893$
Stud. EGARCH	-0.1677	0.0435	0.9926	0.3140					$v = 2.2055$
Stud. GJRGARCH	0.0010	1.0000	0.8274	1.0000					$v = 2.0497$
Stud. APARCH	0.0008	0.3089	0.8737	-0.1853	0.9543				$v = 2.1978$
Stud. IGARCH	0.0001	0.2047	0.7953						$v = 2.6721$
Stud. CSGARCH	0.5064	0.1143	0.8857				0.4980	0.4924	$v = 2.8502$
Stud. TGARCH	0.0007	0.3161	0.8743	-0.1827					$v = 2.1976$
Stud. AVGARCH	0.0008	0.3507	0.8658						$v = 2.1821$
Stud. NGARCH	0.0003	0.7894	0.8739		1.1123				$v = 2.0428$
Stud. NAGARCH	0.0001	1.0000	0.8541			-0.0927			$v = 2.0838$
Stud. ALLGARCH	0.0001	0.2253	0.6314	-0.8511	6.6062	-0.5601			$v = 2.0026$
Skew. Stud. SGARCH	0.0001	0.1180	0.5948						$v = 3.5497, \xi = 0.0004$
Skew. Stud. EGARCH	-0.2028	0.0162	0.9896	0.0945					$v = 3.9796, \xi = 0.0006$
Skew. Stud. GJRGARCH	0.0001	0.3386	0.3700	0.0000					$v = 3.2260, \xi = 0.0001$
Skew. Stud. APARCH	0.0003	0.0808	0.8932	-0.3410	0.9645				$v = 4.0727, \xi = 0.0003$
Skew. Stud. IGARCH	0.0001	0.2921	0.7079						$v = 2.3343, \xi = 0.0001$
Skew. Stud. CSGARCH	0.0001	0.0375	0.8702				0.0000	0.0001	$v = 3.8865, \xi = 0.0001$
Skew. Stud. TGARCH	0.0002	0.0789	0.8933	-0.3328					$v = 4.0686, \xi = 0.0003$
Skew. Stud. AVGARCH	0.0002	0.0840	0.8875						$v = 4.0529, \xi = 0.0003$
Skew. Stud. NGARCH	0.0002	0.0796	0.8883		1.0734				$v = 4.0470, \xi = 0.0002$
Skew. Stud. NAGARCH	0.0001	0.1184	0.5645			0.5645			$v = 3.5318, \xi = 0.0001$
Skew. Stud. ALLGARCH	0.0004	0.0847	0.8913	-0.3573	0.9134	0.0234			$v = 4.0877, \xi = 0.0159$
SU. SGARCH	8.8735	0.4968	0.9896						$\gamma = 10.6714, \delta = 6.8536, \xi = -0.0003, \lambda = 0.0020$
SU. EGARCH	22.9368	0.4946	0.5009	0.5147					$\gamma = 999.9984, \delta = 98.1108, \xi = 0.0000, \lambda = 0.0001$
SU. GJRGARCH	20.8105	0.4785	0.9948	0.4793					$\gamma = 20.1809, \delta = 6.4416, \xi = 0.0000, \lambda = 0.0003$
SU. APARCH	2.1617	0.5021	0.3114	0.0071	0.3114				$\gamma = 4.2674, \delta = 7.8593, \xi = -0.0038, \lambda = 0.0113$
SU. IGARCH	24.1084	0.0464	0.9536						$\gamma = 13.6518, \delta = 7.0365, \xi = -0.0003, \lambda = 0.0016$
SU. CSGARCH	4.1847	0.3139	0.4227				0.0122	0.9949	$\gamma = 15.2123, \delta = 8.0382, \xi = -0.0005, \lambda = 0.0033$
SU. TGARCH	3.3768	0.4288	0.9762	-0.0929					$\gamma = -0.2245, \delta = 4.9552, \xi = -0.0008, \lambda = 0.0012$
SU. AVGARCH	4.8467	0.5398	0.9783						$\gamma = 2.3625, \delta = 5.0970, \xi = -0.0002, \lambda = 0.0009$
SU. NGARCH	2.6288	0.4007	0.7624		0.2651				$\gamma = 3.5915, \delta = 6.2795, \xi = 0.0000, \lambda = 0.0032$
SU. NAGARCH	3.9496	0.6258	0.8947			-0.8754			$\gamma = 4.1239, \delta = 4.7042, \xi = 0.0000, \lambda = 0.0003$
SU. ALLGARCH	2.5239	0.5494	0.5760	0.0925	0.5552	1.8597			$\gamma = 3.8929, \delta = 4.6330, \xi = 0.0000, \lambda = 0.0003$

Table 8: Maximum Likelihood parameter estimates for Ripple

Model	ω	α_1	β_1	γ_1	δ	η_1	ϕ	ρ	Dist.
Norm. SGARCH	0.0004	0.4381	0.5560						
Norm. EGARCH	-1.2262	0.1145	0.8315	0.3654					
Norm. GJRARCH	0.0006	0.8679	0.4351	0.0000					
Norm. APARCH	0.0065	0.4149	0.5606	-0.2976	1.1531				
Norm. IGARCH	0.0004	0.4471	0.5529						
Norm. CSGARCH	0.5156	0.4476	0.5520				0.4877	0.5313	
Norm. TGARCH	0.0100	0.4348	0.5591	0.5591					
Norm. AVGARCH	0.0100	0.4348	0.5591						
Norm. NGARCH	0.0057	0.4441	0.5606		1.1824				
Norm. NAGARCH	0.0005	0.4760	0.4554			-0.3326			
Norm. ALLGARCH	0.0013	0.4374	0.5080	-0.0502	1.6837	-0.2723			
Skew. Norm. SGARCH	0.0010	0.1368	0.8354						$\lambda = -7.6880$
Skew. Norm. EGARCH	-1.9380	0.5354	0.7745	0.4745					$\lambda = 3.0527$
Skew. Norm. GJRARCH	0.0001	0.1812	0.0593	0.1802					$\lambda = 3.0774$
Skew. Norm. APARCH	0.0093	0.6326	0.2704	-0.3098	0.6790				$\lambda = -1.6680$
Skew. Norm. IGARCH	0.0001	0.9690	0.0310						$\lambda = 3.4459$
Skew. Norm. CSGARCH	0.0141	0.5896	0.4095				0.4025	0.1516	$\lambda = 3.0813$
Skew. Norm. TGARCH	0.0101	0.2493	0.3205	-0.1854					$\lambda = -2.0574$
Skew. Norm. AVGARCH	0.0203	0.0711	0.0830						$\lambda = 3.2899$
Skew. Norm. NGARCH	0.0278	0.0037	0.6410		0.7706				$\lambda = -1.4073$
Skew. Norm. NAGARCH	0.0010	0.1223	0.0199			-2.1916			$\lambda = -6.2862$
Skew. Norm. ALLGARCH	0.0768	0.7207	0.3951	-0.3331	0.5787	0.3412			$\lambda = -8.6287$
Stud. SGARCH	0.0003	1.0000	0.5871						$v = 2.3867$
Stud. EGARCH	-0.0818	0.0416	0.9948	0.1462					$v = 2.1836$
Stud. GJRARCH	0.0010	1.0000	0.5357	0.6269					$v = 2.2610$
Stud. APARCH	0.0099	0.5921	0.6324	-0.0976	0.8911				$v = 2.3329$
Stud. IGARCH	0.0002	0.3883	0.6117						$v = 2.9856$
Stud. CSGARCH	0.8501	0.3881	0.6118				0.0942	0.6116	$v = 2.9869$
Stud. TGARCH	0.0072	0.6422	0.6289	-0.0874					$v = 2.3348$
Stud. AVGARCH	0.0069	0.6522	0.6343						$v = 2.3192$
Stud. NGARCH	0.0082	0.6260	0.6359		0.9432				$v = 2.3175$
Stud. NAGARCH	0.0004	0.9999	0.5756			-0.0623			$v = 2.3957$
Stud. ALLGARCH	0.0002	0.7113	0.6984	-0.8704	11.2655	-0.4615			$v = 2.0020$
Skew. Stud. SGARCH	0.0001	0.1720	0.4875						$v = 4.8106, \xi = 0.0001$
Skew. Stud. EGARCH	-0.3359	0.0104	0.9689	0.0893					$v = 4.4911, \xi = 0.0003$
Skew. Stud. GJRARCH	0.0002	0.3627	0.2630	0.0000					$v = 4.5971, \xi = 0.0004$
Skew. Stud. APARCH	0.0064	0.3092	0.4951	-0.2034	0.9355				$v = 3.7380, \xi = 0.0417$
Skew. Stud. IGARCH	0.0001	0.6296	0.3704						$v = 3.0068, \xi = 0.0001$
Skew. Stud. CSGARCH	0.0002	0.1627	0.5344				0.0000	0.0001	$v = 4.8266, \xi = 0.0002$
Skew. Stud. TGARCH	0.0039	0.2443	0.5746	-0.1785					$v = 4.8162, \xi = 0.0004$
Skew. Stud. AVGARCH	0.0033	0.2420	0.5977						$v = 4.8598, \xi = 0.0002$
Skew. Stud. NGARCH	0.0024	0.2359	0.5958		1.0820				$v = 4.8604, \xi = 0.0004$
Skew. Stud. NAGARCH	0.0001	0.1683	0.4735			-0.3818			$v = 4.7483, \xi = 0.0001(1.2402)$
Skew. Stud. ALLGARCH	0.0059	0.2543	0.5706	-0.2070	0.8990	0.0367			$v = 4.8332, \xi = 0.0168$
SU. SGARCH	2.1170	0.5313	0.9806						$\gamma = -1.4592, \delta = 11.5004, \xi = -0.0241, \lambda = 0.0537$
SU. EGARCH	21.8519	0.4998	0.4969	0.5004					$\gamma = 999.9600, \delta = 104.9316, \xi = 0.0000, \lambda = 0.0001$
SU. GJRARCH	3.0595	0.5211	0.9338	0.5170					$\gamma = -0.6170, \delta = 10.5121, \xi = -0.0240, \lambda = 0.0366$
SU. APARCH	4.5308	0.6764	0.2290	0.3364	0.2891				$\gamma = 6.0707, \delta = 18.2087, \xi = -0.0624, \lambda = 0.1103$
SU. IGARCH	2.8846	0.1482	0.8518						$\gamma = -1.0326, \delta = 12.4307, \xi = -0.0499, \lambda = 0.0636$
SU. CSGARCH	25.5560	0.2374	0.4307				0.0043	0.9972	$\gamma = 33.6735, \delta = 8.9112, \xi = -0.0001, \lambda = 0.0004$
SU. TGARCH	22.0918	0.0671	0.7087	-0.0355					$\gamma = 41.6505, \delta = 10.5539, \xi = 0.0000, \lambda = 0.0007$
SU. AVGARCH	14.0559	0.6244	0.7868						$\gamma = 26.9361, \delta = 8.8022, \xi = 0.0000, \lambda = 0.0008$
SU. NGARCH	4.2616	0.6797	0.2377		0.3563				$\gamma = 6.7754, \delta = 20.0138, \xi = 0.0034, \lambda = 0.1471$
SU. NAGARCH	1.8115	0.9360	0.9360			0.7370			$\gamma = -0.1238, \delta = 6.4069, \xi = -0.0013, \lambda = 0.0035$
SU. ALLGARCH	1.5794	0.5070	0.4544	0.0423	2.6068	1.8021			$\gamma = 6.9097, \delta = 18.8946, \xi = -0.0083, \lambda = 0.1253$

Table 9: Loglikelihood, Information Criteria and Logarithmic Scoring Rule for Bitcoin

Model	Loglikelihood	AIC	BIC	CAIC	AICc	HQC	LSR
Norm. SGARCH	2.9290e+03	-5.8521e+03	-5.8362e+03	-5.8332e+03	-5.8521e+03	-5.8462e+03	1.8886
Norm. EGARCH	2.9528e+03	-5.8977e+03	-5.8764e+03	-5.8724e+03	-5.8977e+03	-5.8898e+03	1.8154
Norm. GJRGARCH	2.8889e+03	-5.7698e+03	-5.7486e+03	-5.7446e+03	-5.7698e+03	-5.7619e+03	1.8770
Norm. APARCH	2.9534e+03	-5.8969e+03	-5.8703e+03	-5.8653e+03	-5.8968e+03	-5.8870e+03	1.8230
Norm. IGARCH	2.9223e+03	-5.8385e+03	-5.8226e+03	-5.8196e+03	-5.8385e+03	-5.8326e+03	1.8785
Norm. CSGARCH	2.9429e+03	-5.8758e+03	-5.8492e+03	-5.8442e+03	-5.8758e+03	-5.8659e+03	1.8209
Norm. TGARCH	2.9528e+03	-5.8977e+03	-5.8764e+03	-5.8724e+03	-5.8977e+03	-5.8898e+03	1.8232
Norm. AVGARCH	2.9525e+03	-5.8989e+03	-5.8830e+03	-5.8800e+03	-5.8989e+03	-5.8930e+03	1.8214
Norm. NGARCH	2.9197e+03	-5.8315e+03	-5.8102e+03	-5.8062e+03	-5.8314e+03	-5.8236e+03	1.8798
Norm. NAGARCH	2.9297e+03	-5.8513e+03	-5.8301e+03	-5.8261e+03	-5.8513e+03	-5.8434e+03	1.8864
Norm. ALLGARCH	2.9533e+03	-5.8946e+03	-5.8627e+03	-5.8567e+03	-5.8945e+03	-5.8827e+03	1.8358
Skew. Norm. SGARCH	1.9709e+04	-3.9410e+04	-3.9388e+04	-3.9384e+04	-3.9410e+04	-3.9402e+04	14.5289
Skew. Norm. EGARCH	1.2069e+04	-2.4128e+04	-2.4102e+04	-2.4097e+04	-2.4128e+04	-2.4118e+04	10.0363
Skew. Norm. GJRGARCH	2.0080e+04	-4.0150e+04	-4.0123e+04	-4.0118e+04	-4.0150e+04	-4.0140e+04	11.6975
Skew. Norm. APARCH	1.5295e+04	-3.0578e+04	-3.0546e+04	-3.0540e+04	-3.0578e+04	-3.0566e+04	11.2045
Skew. Norm. IGARCH	1.1891e+04	-2.3774e+04	-2.3753e+04	-2.3749e+04	-2.3774e+04	-2.3766e+04	7.5115
Skew. Norm. CSGARCH	1.1121e+04	-2.2231e+04	-2.2199e+04	-2.2193e+04	-2.2231e+04	-2.2219e+04	7.2180
Skew. Norm. TGARCH	1.9752e+04	-3.9493e+04	-3.9467e+04	-3.9462e+04	-3.9493e+04	-3.9483e+04	12.8514
Skew. Norm. AVGARCH	2.0298e+04	-4.0589e+04	-4.0568e+04	-4.0564e+04	-4.0589e+04	-4.0581e+04	12.9464
Skew. Norm. NGARCH	1.5111e+04	-3.0211e+04	-3.0185e+04	-3.0180e+04	-3.0211e+04	-3.0202e+04	7.6818
Skew. Norm. NAGARCH	2.1705e+04	-4.3400e+04	-4.3374e+04	-4.3369e+04	-4.3400e+04	-4.3390e+04	13.9837
Skew. Norm. ALLGARCH	1.9135e+04	-3.8256e+04	-3.8218e+04	-3.8211e+04	-3.8256e+04	-3.8242e+04	11.5343
Stud. SGARCH	3.1298e+03	-6.2516e+03	-6.2304e+03	-6.2264e+03	-6.2516e+03	-6.2437e+03	2.0655
Stud. EGARCH	3.1278e+03	-6.2457e+03	-6.2191e+03	-6.2141e+03	-6.2456e+03	-6.2358e+03	2.0487
Stud. GJRGARCH	3.1302e+03	-6.2505e+03	-6.2239e+03	-6.2189e+03	-6.2504e+03	-6.2406e+03	2.0658
Stud. APARCH	3.1485e+03	-6.2851e+03	-6.2532e+03	-6.2472e+03	-6.2850e+03	-6.2732e+03	2.0781
Stud. IGARCH	3.1134e+03	-6.2188e+03	-6.1976e+03	-6.1936e+03	-6.2188e+03	-6.2109e+03	2.0551
Stud. CSGARCH	3.1306e+03	-6.2493e+03	-6.2174e+03	-6.2114e+03	-6.2492e+03	-6.2374e+03	2.0563
Stud. TGARCH	3.1478e+03	-6.2856e+03	-6.2590e+03	-6.2540e+03	-6.2856e+03	-6.2757e+03	2.0769
Stud. AVGARCH	3.1466e+03	-6.2851e+03	-6.2639e+03	-6.2599e+03	-6.2851e+03	-6.2772e+03	2.0772
Stud. NGARCH	3.1471e+03	-6.2842e+03	-6.2577e+03	-6.2527e+03	-6.2842e+03	-6.2743e+03	2.0790
Stud. NAGARCH	3.1305e+03	-6.2510e+03	-6.2245e+03	-6.2195e+03	-6.2510e+03	-6.2411e+03	2.0672
Stud. ALLGARCH	3.1389e+03	-6.2639e+03	-6.2267e+03	-6.2197e+03	-6.2638e+03	-6.2500e+03	2.0805
Skew. Stud. SGARCH	3.7346e+03	-7.4591e+03	-7.4326e+03	-7.4276e+03	-7.4591e+03	-7.4492e+03	2.4005
Skew. Stud. EGARCH	3.8033e+03	-7.5947e+03	-7.5628e+03	-7.5568e+03	-7.5946e+03	-7.5828e+03	2.4189
Skew. Stud. GJRGARCH	3.7076e+03	-7.4032e+03	-7.3713e+03	-7.3653e+03	-7.4032e+03	-7.3913e+03	2.3774
Skew. Stud. APARCH	3.8299e+03	-7.6459e+03	-7.6087e+03	-7.6017e+03	-7.6458e+03	-7.6320e+03	3.2158
Skew. Stud. IGARCH	3.7182e+03	-7.4265e+03	-7.3999e+03	-7.3949e+03	-7.4264e+03	-7.4166e+03	2.3746
Skew. Stud. CSGARCH	3.3229e+03	-6.6318e+03	-6.5946e+03	-6.5876e+03	-6.6317e+03	-6.6179e+03	2.1070
Skew. Stud. TGARCH	3.8299e+03	-7.6478e+03	-7.6159e+03	-7.6099e+03	-7.6478e+03	-7.6359e+03	2.4481
Skew. Stud. AVGARCH	3.8263e+03	-7.6426e+03	-7.6160e+03	-7.6110e+03	-7.6425e+03	-7.6327e+03	2.4494
Skew. Stud. NGARCH	3.7114e+03	-7.4107e+03	-7.3788e+03	-7.3728e+03	-7.4107e+03	-7.3988e+03	2.3727
Skew. Stud. NAGARCH	3.7356e+03	-7.4593e+03	-7.4274e+03	-7.4214e+03	-7.4592e+03	-7.4474e+03	2.4037
Skew. Stud. ALLGARCH	3.8276e+03	-7.6392e+03	-7.5967e+03	-7.5887e+03	-7.6391e+03	-7.6234e+03	2.4563
SU. SGARCH	6.6621e+03	-1.3310e+04	-1.3273e+04	-1.3266e+04	-1.3310e+04	-1.3296e+04	5.0146
SU. EGARCH	2.0622e+04	-4.1229e+04	-4.1186e+04	-4.1178e+04	-4.1229e+04	-4.1213e+04	13.7354
SU. GJRGARCH	8.6923e+03	-1.7369e+04	-1.7326e+04	-1.7318e+04	-1.7368e+04	-1.7353e+04	6.4876
SU. APARCH	9.2975e+03	-1.8577e+04	-1.8529e+04	-1.8520e+04	-1.8577e+04	-1.8559e+04	6.7307
SU. IGARCH	5.5456e+03	-1.1077e+04	-1.1040e+04	-1.1033e+04	-1.1077e+04	-1.1063e+04	4.2250
SU. CSGARCH	1.3781e+04	-2.7545e+04	-2.7497e+04	-2.7488e+04	-2.7545e+04	-2.7527e+04	9.8958
SU. TGARCH	1.4926e+04	-2.9836e+04	-2.9793e+04	-2.9785e+04	-2.9836e+04	-2.9820e+04	10.5517
SU. AVGARCH	1.6059e+04	-3.2104e+04	-3.2067e+04	-3.2060e+04	-3.2104e+04	-3.2090e+04	11.2118
SU. NGARCH	9.2378e+03	-1.8460e+04	-1.8417e+04	-1.8409e+04	-1.8459e+04	-1.8444e+04	6.6932
SU. NAGARCH	7.1717e+03	-1.4327e+04	-1.4285e+04	-1.4277e+04	-1.4327e+04	-1.4312e+04	5.2995
SU. ALLGARCH	9.8591e+03	-1.9698e+04	-1.9645e+04	-1.9635e+04	-1.9698e+04	-1.9678e+04	7.1120

Table 10: Loglikelihood, Information Criteria and Logarithmic Scoring Rule for Litecoin

Model	Loglikelihood	AIC	BIC	CAIC	AICc	HQC	LSR
Norm. SGARCH	2.3191e+03	-4.6322e+03	-4.6162e+03	-4.6132e+03	-4.6321e+03	-4.6262e+03	1.5433
Norm. EGARCH	2.3371e+03	-4.6661e+03	-4.6449e+03	-4.6409e+03	-4.6661e+03	-4.6582e+03	1.5294
Norm. GJRGARCH	2.2566e+03	-4.5052e+03	-4.4839e+03	-4.4799e+03	-4.5052e+03	-4.4973e+03	1.5336
Norm. APARCH	2.3479e+03	-4.6858e+03	-4.6592e+03	-4.6542e+03	-4.6857e+03	-4.6759e+03	1.5158
Norm. IGARCH	2.3037e+03	-4.6014e+03	-4.5855e+03	-4.5825e+03	-4.6014e+03	-4.5955e+03	1.5195
Norm. CSGARCH	2.3038e+03	-4.5977e+03	-4.5711e+03	-4.5661e+03	-4.5976e+03	-4.5878e+03	1.5195
Norm. TGARCH	2.3182e+03	-4.6284e+03	-4.6071e+03	-4.6031e+03	-4.6283e+03	-4.6204e+03	1.5295
Norm. AVGARCH	2.3182e+03	-4.6304e+03	-4.6144e+03	-4.6114e+03	-4.6303e+03	-4.6244e+03	1.5408
Norm. NGARCH	2.3200e+03	-4.6320e+03	-4.6108e+03	-4.6068e+03	-4.6320e+03	-4.6241e+03	1.5437
Norm. NAGARCH	2.3435e+03	-4.6790e+03	-4.6577e+03	-4.6537e+03	-4.6790e+03	-4.6711e+03	1.5149
Norm. ALLGARCH	2.2976e+03	-4.5832e+03	-4.5513e+03	-4.5453e+03	-4.5831e+03	-4.5713e+03	1.5231
Skew. Norm. SGARCH	1.0746e+04	-2.1484e+04	-2.1463e+04	-2.1459e+04	-2.1484e+04	-2.1476e+04	7.3734
Skew. Norm. EGARCH	1.2773e+04	-2.5535e+04	-2.5508e+04	-2.5503e+04	-2.5535e+04	-2.5525e+04	11.1032
Skew. Norm. GJRGARCH	1.7446e+04	-3.4882e+04	-3.4855e+04	-3.4850e+04	-3.4882e+04	-3.4872e+04	10.1766
Skew. Norm. APARCH	9.9759e+03	-1.9940e+04	-1.9908e+04	-1.9902e+04	-1.9940e+04	-1.9928e+04	5.5720
Skew. Norm. IGARCH	1.2220e+04	-2.4432e+04	-2.4410e+04	-2.4406e+04	-2.4432e+04	-2.4424e+04	8.6447
Skew. Norm. CSGARCH	1.0747e+04	-2.1482e+04	-2.1450e+04	-2.1444e+04	-2.1482e+04	-2.1470e+04	7.2783
Skew. Norm. TGARCH	8.8656e+03	-1.7721e+04	-1.7695e+04	-1.7690e+04	-1.7721e+04	-1.7711e+04	5.2919
Skew. Norm. AVGARCH	1.3522e+04	-2.7035e+04	-2.7014e+04	-2.7010e+04	-2.7035e+04	-2.7027e+04	8.7115
Skew. Norm. NGARCH	8.2148e+03	-1.6420e+04	-1.6393e+04	-1.6388e+04	-1.6420e+04	-1.6410e+04	4.3434
Skew. Norm. NAGARCH	9.7532e+03	-1.9496e+04	-1.9470e+04	-1.9465e+04	-1.9496e+04	-1.9486e+04	5.0535
Skew. Norm. ALLGARCH	9.7135e+03	-1.9413e+04	-1.9376e+04	-1.9369e+04	-1.9413e+04	-1.9399e+04	5.4199
Stud. SGARCH	2.7998e+03	-5.5916e+03	-5.5703e+03	-5.5663e+03	-5.5915e+03	-5.5837e+03	1.6359
Stud. EGARCH	2.7872e+03	-5.5643e+03	-5.5378e+03	-5.5328e+03	-5.5643e+03	-5.5544e+03	1.6349
Stud. GJRGARCH	2.7639e+03	-5.5179e+03	-5.4913e+03	-5.4863e+03	-5.5178e+03	-5.5080e+03	1.6333
Stud. APARCH	2.8144e+03	-5.6168e+03	-5.5849e+03	-5.5789e+03	-5.6167e+03	-5.6049e+03	1.6428
Stud. IGARCH	2.7743e+03	-5.5405e+03	-5.5193e+03	-5.5153e+03	-5.5405e+03	-5.5326e+03	1.6310
Stud. CSGARCH	2.7882e+03	-5.5644e+03	-5.5325e+03	-5.5265e+03	-5.5643e+03	-5.5525e+03	1.6344
Stud. TGARCH	2.8143e+03	-5.6187e+03	-5.5921e+03	-5.5871e+03	-5.6186e+03	-5.6088e+03	1.6427
Stud. AVGARCH	2.8120e+03	-5.6160e+03	-5.5948e+03	-5.5908e+03	-5.6160e+03	-5.6081e+03	1.6461
Stud. NGARCH	2.8090e+03	-5.6080e+03	-5.5814e+03	-5.5764e+03	-5.6079e+03	-5.5981e+03	1.6439
Stud. NAGARCH	2.8013e+03	-5.5926e+03	-5.5661e+03	-5.5611e+03	-5.5926e+03	-5.5827e+03	1.6315
Stud. ALLGARCH	2.7393e+03	-5.4646e+03	-5.4274e+03	-5.4204e+03	-5.4646e+03	-5.4508e+03	1.5686
Skew. Stud. SGARCH	3.2129e+03	-6.4159e+03	-6.3893e+03	-6.3843e+03	-6.4159e+03	-6.4060e+03	1.9703
Skew. Stud. EGARCH	3.2730e+03	-6.5341e+03	-6.5022e+03	-6.4962e+03	-6.5222e+03	-6.5222e+03	1.9714
Skew. Stud. GJRGARCH	3.1763e+03	-6.3405e+03	-6.3086e+03	-6.3026e+03	-6.3405e+03	-6.3286e+03	1.9232
Skew. Stud. APARCH	3.3111e+03	-6.6083e+03	-6.5711e+03	-6.5641e+03	-6.6082e+03	-6.5944e+03	2.7785
Skew. Stud. IGARCH	3.1802e+03	-6.3504e+03	-6.3238e+03	-6.3188e+03	-6.3503e+03	-6.3405e+03	1.9049
Skew. Stud. CSGARCH	3.2845e+03	-6.5550e+03	-6.5178e+03	-6.5108e+03	-6.5549e+03	-6.5411e+03	1.9752
Skew. Stud. TGARCH	3.3111e+03	-6.6102e+03	-6.5783e+03	-6.5723e+03	-6.6102e+03	-6.5983e+03	1.9744
Skew. Stud. AVGARCH	3.3007e+03	-6.5913e+03	-6.5648e+03	-6.5598e+03	-6.5913e+03	-6.5814e+03	1.9846
Skew. Stud. NGARCH	3.3008e+03	-6.5896e+03	-6.5577e+03	-6.5517e+03	-6.5895e+03	-6.5777e+03	1.9842
Skew. Stud. NAGARCH	3.2208e+03	-6.4296e+03	-6.3977e+03	-6.3917e+03	-6.4296e+03	-6.4178e+03	1.9657
Skew. Stud. ALLGARCH	3.3109e+03	-6.6058e+03	-6.5633e+03	-6.5553e+03	-6.6057e+03	-6.5900e+03	1.9752
SU. SGARCH	1.0935e+04	-2.1856e+04	-2.1819e+04	-2.1812e+04	-2.1856e+04	-2.1842e+04	7.9329
SU. EGARCH	2.0674e+04	-4.1332e+04	-4.1289e+04	-4.1281e+04	-4.1332e+04	-4.1316e+04	13.7735
SU. GJRGARCH	1.3199e+04	-2.6383e+04	-2.6340e+04	-2.6332e+04	-2.6383e+04	-2.6367e+04	9.3560
SU. APARCH	8.7494e+03	-1.7481e+04	-1.7433e+04	-1.7424e+04	-1.7481e+04	-1.7463e+04	6.4275
SU. IGARCH	1.1165e+04	-2.2316e+04	-2.2279e+04	-2.2272e+04	-2.2316e+04	-2.2302e+04	8.0348
SU. CSGARCH	9.6532e+03	-1.9288e+04	-1.9241e+04	-1.9232e+04	-1.9288e+04	-1.9271e+04	7.2780
SU. TGARCH	1.1068e+04	-2.2120e+04	-2.2077e+04	-2.2069e+04	-2.2119e+04	-2.2104e+04	8.0913
SU. AVGARCH	1.1946e+04	-2.3878e+04	-2.3840e+04	-2.3833e+04	-2.3878e+04	-2.3864e+04	8.6085
SU. NGARCH	1.0618e+04	-2.1220e+04	-2.1177e+04	-2.1169e+04	-2.1220e+04	-2.1204e+04	7.5790
SU. NAGARCH	1.3096e+04	-2.6176e+04	-2.6133e+04	-2.6125e+04	-2.6176e+04	-2.6160e+04	9.6458
SU. ALLGARCH	1.3658e+04	-2.7297e+04	-2.7244e+04	-2.7234e+04	-2.7297e+04	-2.7277e+04	9.7045

Table 11: Loglikelihood, Information Criteria and Logarithmic Scoring Rule for Ripple

Model	Loglikelihood	AIC	BIC	CAIC	AICc	HQC	LSR
Norm. SGARCH	2.2758e+03	-4.5457e+03	-4.5297e+03	-4.5267e+03	-4.5457e+03	-4.5397e+03	1.8123
Norm. EGARCH	2.2540e+03	-4.4999e+03	-4.4787e+03	-4.4747e+03	-4.4999e+03	-4.4920e+03	1.8080
Norm. GJRGARCH	2.2867e+03	-4.5653e+03	-4.5441e+03	-4.5401e+03	-4.5653e+03	-4.5574e+03	1.8058
Norm. APARCH	2.2934e+03	-4.5769e+03	-4.5503e+03	-4.5453e+03	-4.5769e+03	-4.5670e+03	1.8108
Norm. IGARCH	2.2758e+03	-4.5457e+03	-4.5297e+03	-4.5267e+03	-4.5456e+03	-4.5397e+03	1.8116
Norm. CSGARCH	2.2760e+03	-4.5419e+03	-4.5154e+03	-4.5104e+03	-4.5419e+03	-4.5320e+03	1.8117
Norm. TGARCH	2.2688e+03	-4.5297e+03	-4.5084e+03	-4.5044e+03	-4.5297e+03	-4.5218e+03	1.6770
Norm. AVGARCH	2.2688e+03	-4.5317e+03	-4.5157e+03	-4.5127e+03	-4.5317e+03	-4.5257e+03	1.7963
Norm. NGARCH	2.2714e+03	-4.5347e+03	-4.5135e+03	-4.5095e+03	-4.5347e+03	-4.5268e+03	1.8009
Norm. NAGARCH	2.2956e+03	-4.5833e+03	-4.5620e+03	-4.5580e+03	-4.5832e+03	-4.5753e+03	1.8143
Norm. ALLGARCH	2.2960e+03	-4.5800e+03	-4.5481e+03	-4.5421e+03	-4.5800e+03	-4.5681e+03	1.8153
Skew. Norm. SGARCH	1.5003e+04	-2.9998e+04	-2.9977e+04	-2.9973e+04	-2.9998e+04	-2.9991e+04	6.0802
Skew. Norm. EGARCH	1.4527e+04	-2.9045e+04	-2.9018e+04	-2.9013e+04	-2.9045e+04	-2.9035e+04	8.7672
Skew. Norm. GJRGARCH	2.5375e+04	-5.0739e+04	-5.0713e+04	-5.0708e+04	-5.0739e+04	-5.0729e+04	15.2325
Skew. Norm. APARCH	9.2649e+03	-1.8518e+04	-1.8518e+04	-1.8480e+04	-1.8518e+04	-1.8506e+04	4.7738
Skew. Norm. IGARCH	2.0233e+04	-4.0457e+04	-4.0436e+04	-4.0432e+04	-4.0457e+04	-4.0450e+04	11.7637
Skew. Norm. CSGARCH	1.3229e+04	-2.6447e+04	-2.6415e+04	-2.6409e+04	-2.6447e+04	-2.6435e+04	11.1145
Skew. Norm. TGARCH	1.1386e+04	-2.2762e+04	-2.2735e+04	-2.2730e+04	-2.2762e+04	-2.2752e+04	5.7813
Skew. Norm. AVGARCH	1.8478e+04	-3.6947e+04	-3.6926e+04	-3.6922e+04	-3.6947e+04	-3.6939e+04	10.2300
Skew. Norm. NGARCH	6.8480e+03	-1.3686e+04	-1.3659e+04	-1.3654e+04	-1.3686e+04	-1.3676e+04	3.8069
Skew. Norm. NAGARCH	2.1856e+04	-4.3701e+04	-4.3675e+04	-4.3670e+04	-4.3701e+04	-4.3691e+04	9.5008
Skew. Norm. ALLGARCH	1.3131e+04	-2.6247e+04	-2.6210e+04	-2.6203e+04	-2.6247e+04	-2.6234e+04	4.9832
Stud. SGARCH	2.5803e+03	-5.1526e+03	-5.1313e+03	-5.1273e+03	-5.1526e+03	-5.1447e+03	1.9509
Stud. EGARCH	2.5304e+03	-5.0509e+03	-5.0243e+03	-5.0193e+03	-5.0508e+03	-5.0410e+03	1.9341
Stud. GJRGARCH	2.5711e+03	-5.1322e+03	-5.1057e+03	-5.1007e+03	-5.1322e+03	-5.1223e+03	1.9514
Stud. APARCH	2.5891e+03	-5.1663e+03	-5.1344e+03	-5.1284e+03	-5.1662e+03	-5.1544e+03	1.9525
Stud. IGARCH	2.5696e+03	-5.1312e+03	-5.1100e+03	-5.1060e+03	-5.1312e+03	-5.1233e+03	1.9543
Stud. CSGARCH	2.5696e+03	-5.1273e+03	-5.0954e+03	-5.0894e+03	-5.1154e+03	-5.1154e+03	1.9535
Stud. TGARCH	2.5890e+03	-5.1680e+03	-5.1414e+03	-5.1364e+03	-5.1581e+03	-5.1581e+03	1.9525
Stud. AVGARCH	2.5880e+03	-5.1680e+03	-5.1468e+03	-5.1428e+03	-5.1680e+03	-5.1601e+03	1.9510
Stud. NGARCH	2.5880e+03	-5.1661e+03	-5.1395e+03	-5.1345e+03	-5.1661e+03	-5.1562e+03	1.9509
Stud. NAGARCH	2.5811e+03	-5.1521e+03	-5.1256e+03	-5.1206e+03	-5.1521e+03	-5.1422e+03	1.9515
Stud. ALLGARCH	2.5286e+03	-5.0432e+03	-5.0060e+03	-4.9990e+03	-5.0432e+03	-5.0294e+03	1.9357
Skew. Stud. SGARCH	3.0205e+03	-6.0309e+03	-6.0043e+03	-5.9993e+03	-6.0309e+03	-6.0210e+03	2.2396
Skew. Stud. EGARCH	2.9389e+03	-5.8659e+03	-5.8340e+03	-5.8280e+03	-5.8658e+03	-5.8540e+03	2.2161
Skew. Stud. GJRGARCH	3.0113e+03	-6.0106e+03	-5.9787e+03	-5.9727e+03	-6.0106e+03	-5.9987e+03	2.2251
Skew. Stud. APARCH	3.0226e+03	-6.0313e+03	-5.9941e+03	-5.9871e+03	-6.0312e+03	-6.0174e+03	2.9170
Skew. Stud. IGARCH	2.9608e+03	-5.9115e+03	-5.8850e+03	-5.8800e+03	-5.9115e+03	-5.9017e+03	2.1719
Skew. Stud. CSGARCH	3.0210e+03	-6.0279e+03	-5.9907e+03	-5.9837e+03	-6.0279e+03	-6.0141e+03	2.2392
Skew. Stud. TGARCH	3.0327e+03	-6.0535e+03	-6.0216e+03	-6.0156e+03	-6.0534e+03	-6.0416e+03	2.2469
Skew. Stud. AVGARCH	3.0274e+03	-6.0448e+03	-6.0182e+03	-6.0132e+03	-6.0447e+03	-6.0349e+03	2.2381
Skew. Stud. NGARCH	3.0275e+03	-6.0429e+03	-6.0111e+03	-6.0051e+03	-6.0429e+03	-6.0311e+03	2.2386
Skew. Stud. NAGARCH	3.0267e+03	-6.0414e+03	-6.0095e+03	-6.0035e+03	-6.0413e+03	-6.0295e+03	2.2480
Skew. Stud. ALLGARCH	3.0327e+03	-6.0494e+03	-6.0069e+03	-5.9989e+03	-6.0493e+03	-6.0335e+03	2.2455
SU. SGARCH	7.0735e+03	-1.4133e+04	-1.4096e+04	-1.4089e+04	-1.4133e+04	-1.4119e+04	5.2711
SU. EGARCH	2.0776e+04	-4.1536e+04	-4.1494e+04	-4.1486e+04	-4.1536e+04	-4.1521e+04	13.8279
SU. GJRGARCH	7.2214e+03	-1.4427e+04	-1.4384e+04	-1.4376e+04	-1.4427e+04	-1.4411e+04	5.4783
SU. APARCH	6.7053e+03	-1.3393e+04	-1.3345e+04	-1.3336e+04	-1.3392e+04	-1.3375e+04	4.9676
SU. IGARCH	6.6218e+03	-1.3230e+04	-1.3192e+04	-1.3185e+04	-1.3230e+04	-1.3216e+04	5.0373
SU. CSGARCH	1.2605e+04	-2.5192e+04	-2.5144e+04	-2.5135e+04	-2.5192e+04	-2.5174e+04	9.3552
SU. TGARCH	1.3340e+04	-2.6664e+04	-2.6622e+04	-2.6614e+04	-2.6664e+04	-2.6649e+04	9.4719
SU. AVGARCH	1.2834e+04	-2.5654e+04	-2.5617e+04	-2.5610e+04	-2.5654e+04	-2.5640e+04	9.1340
SU. NGARCH	6.6232e+03	-1.3230e+04	-1.3188e+04	-1.3180e+04	-1.3215e+04	-1.3215e+04	4.9128
SU. NAGARCH	9.9269e+03	-1.9838e+04	-1.9795e+04	-1.9787e+04	-1.9838e+04	-1.9822e+04	7.4387
SU. ALLGARCH	6.7747e+03	-1.3529e+04	-1.3476e+04	-1.3466e+04	-1.3529e+04	-1.3510e+04	5.0139

Table 12: Diebold-Mariano test: model comparison for Bitcoin

Model 1	Model 2	Normal	Skew. Normal	Student t	Skew. Student t	SU
SGARCH	EGARCH	1.2797	1.5599	0.9324	-0.6278	-228.5644
SGARCH	GJRGARCH	0.7175	1.0187	-0.2143	2.4856	-323.1359
SGARCH	APARCH	1.1527	3.4808	-0.9705	-11.4520	-51.8831
SGARCH	IGARCH	1.2711	2.7529	1.5282	2.4357	38.8243
SGARCH	CSGARCH	1.4520	2.9068	0.9378	3.9923	-130.2591
SGARCH	TGARCH	2.0025	-0.8111	-1.0591	-1.8257	-184.8664
SGARCH	AVGARCH	1.2328	0.5282	-1.2321	-2.0111	-189.3971
SGARCH	NGARCH	0.3287	2.4402	-1.2300	1.2611	-54.5308
SGARCH	NAGARCH	0.7412	0.6841	-1.0616	-1.3350	-8.6854
SGARCH	ALLGARCH	0.9653	1.7325	-1.4686	-2.1289	-63.9463
EGARCH	GJRGARCH	-1.0054	-1.2841	-0.9626	1.2932	203.4819
EGARCH	APARCH	-0.6403	-0.4389	-2.3185	-9.5969	365.5007
EGARCH	IGARCH	-1.1559	1.7060	-0.3360	1.5270	412.4278
EGARCH	CSGARCH	-0.3415	2.0247	-0.4674	4.6839	107.5977
EGARCH	TGARCH	-0.0964	-1.9475	-2.1250	-2.0242	156.7862
EGARCH	AVGARCH	-0.5795	-2.5674	-1.9858	-1.9380	131.3747
EGARCH	NGARCH	-1.5738	0.6863	-2.1771	1.7974	363.1106
EGARCH	NAGARCH	-1.2527	-1.5099	-1.0435	0.5206	439.2596
EGARCH	ALLGARCH	-1.3393	-0.6746	-2.1577	-2.3843	344.8790
GJRGARCH	APARCH	0.8874	0.1860	-0.9652	-11.5841	-7.6592
GJRGARCH	IGARCH	-0.1187	4.1023	1.6009	0.2249	131.0982
GJRGARCH	CSGARCH	1.1025	4.1970	0.9580	3.7313	-98.4759
GJRGARCH	TGARCH	1.7378	-1.4034	-1.0514	-2.5460	-152.3855
GJRGARCH	AVGARCH	0.9379	-2.0980	-1.1979	-2.7738	-158.9802
GJRGARCH	NGARCH	-0.0982	1.1820	-1.2050	0.2074	-7.4134
GJRGARCH	NAGARCH	-0.6403	-0.9157	-1.5664	-2.6014	39.8697
GJRGARCH	ALLGARCH	0.7107	0.0767	-1.4604	-2.8547	-20.9537
APARCH	IGARCH	-1.0114	1.5404	1.5050	11.5528	198.5361
APARCH	CSGARCH	0.0933	1.6976	1.9540	11.5528	-100.6955
APARCH	TGARCH	-0.0023	-2.6702	0.4142	9.2361	-651.2645
APARCH	AVGARCH	0.3341	-0.6072	0.1917	9.4010	-4.3429e+04
APARCH	NGARCH	-1.4664	1.4485	-0.2588	10.8390	9.9303
APARCH	NAGARCH	-1.1235	-2.5786	0.8657	11.3458	5.9297e+04
APARCH	ALLGARCH	-1.9262	-0.2009	-0.4024	9.1018	-1.6591e+04
IGARCH	CSGARCH	1.3178	1.4031	-0.1148	3.5447	-193.5040
IGARCH	TGARCH	1.6173	-3.2312	-1.6443	-2.9967	-625.8528
IGARCH	AVGARCH	1.0851	-5.2574	-1.8211	-3.2919	-560.1738
IGARCH	NGARCH	-0.0526	-0.0525	-1.7746	0.1328	-235.0433
IGARCH	NAGARCH	-1.0482	-0.0525	-1.7108	-2.6880	-85.5578
IGARCH	ALLGARCH	0.8135	-2.2779	-1.7728	-3.2276	-229.8922
CSGARCH	TGARCH	-0.0320	-3.4177	-2.3834	-5.2529	-24.2251
CSGARCH	AVGARCH	-0.0260	-5.2836	-2.6782	-5.3003	-41.7707
CSGARCH	NGARCH	-1.6285	-0.1434	-2.3998	-3.5178	105.8096
CSGARCH	NAGARCH	-1.4300	-3.0927	-1.0948	-4.0246	145.4738
CSGARCH	ALLGARCH	-0.6450	-2.5188	-1.8883	-5.3556	88.1064
TGARCH	AVGARCH	0.0222	0.8940	-0.1188	-0.2624	-114.0381
TGARCH	NGARCH	-1.1090	3.3109	-0.8420	3.8726	940.8129
TGARCH	NAGARCH	-1.9869	1.1700	0.9302	1.7183	888.5985
TGARCH	ALLGARCH	-0.1614	2.9820	-0.6168	-1.3262	581.9380
AVGARCH	NGARCH	-1.5571	1.4744	-0.8835	4.2599	2.0735e+03
AVGARCH	NAGARCH	-1.2006	-0.3826	1.0550	1.8730	4.7472e+04
AVGARCH	ALLGARCH	-1.5369	0.6126	-0.5310	-0.8709	3.2919e+04
NGARCH	NAGARCH	-0.2465	-2.3263	1.0827	-1.4073	607.8406
NGARCH	ALLGARCH	1.2544	-1.6296	-0.2646	-3.9844	-182.6773
NAGARCH	ALLGARCH	0.9340	1.8593	-1.3615	-2.0317	-3.9770e+15

Table 13: Diebold-Mariano test: model comparison for Litecoin

Model 1	Model 2	Normal	Skew. Normal	Student t	Skew. Student t	SU
SGARCH	EGARCH	1.1006	-3.0768	0.0852	-0.0493	-234.6491
SGARCH	GJRGARCH	0.5768	-3.8100	0.4545	3.8200	-80.1886
SGARCH	APARCH	1.3549	1.4521	-1.1079	-10.7660	197.7604
SGARCH	IGARCH	3.7414	-1.6266	0.8188	6.2791	-10.0020
SGARCH	CSGARCH	1.5440	0.5756	0.2314	-0.3812	7.2840
SGARCH	TGARCH	0.7382	1.8561	-1.1387	-0.2319	-21.8542
SGARCH	AVGARCH	0.2457	-3.0239	-1.8005	-0.9221	-64.0130
SGARCH	NGARCH	-0.0609	3.3445	-1.4853	-0.9121	20.5369
SGARCH	NAGARCH	2.1812	2.1666	1.6870	0.7384	-99.4037
SGARCH	ALLGARCH	1.0063	1.9487	3.5018	-0.2765	-102.8149
EGARCH	GJRGARCH	-0.2412	1.9743	0.1205	1.5851	127.6495
EGARCH	APARCH	1.2526	2.8679	-0.7733	-10.4706	331.9757
EGARCH	IGARCH	0.7982	2.1640	0.2630	2.7249	175.0887
EGARCH	CSGARCH	-0.7403	3.0498	0.0395	-0.2383	65.6450
EGARCH	TGARCH	-0.0033	3.1125	-0.7672	-0.2348	247.8656
EGARCH	AVGARCH	-1.1972	2.2197	-1.0246	-0.9711	273.1682
EGARCH	NGARCH	-1.4639	3.8934	-0.8279	-0.9453	357.7508
EGARCH	NAGARCH	1.0820	3.2989	0.2883	0.2518	238.3880
EGARCH	ALLGARCH	0.3972	3.1033	3.8918	-0.2967	234.9932
GJRGARCH	APARCH	0.8084	2.6291	-0.9763	-11.3115	119.1853
GJRGARCH	IGARCH	0.9643	1.5102	0.3408	1.1213	82.0409
GJRGARCH	CSGARCH	-0.3057	3.9149	-0.1153	-2.3032	23.2604
GJRGARCH	TGARCH	0.2407	3.0610	-0.9836	-2.0303	52.3001
GJRGARCH	AVGARCH	-0.5735	1.3300	-1.4826	-2.6976	30.1468
GJRGARCH	NGARCH	-0.7416	4.4920	-1.2364	-2.6843	61.9468
GJRGARCH	NAGARCH	0.7270	3.3815	0.2553	-2.8758	-10.1028
GJRGARCH	ALLGARCH	0.6710	3.1492	3.5093	-2.0732	-12.1519
APARCH	IGARCH	-0.1980	-1.8135	1.2172	11.6216	-99.1314
APARCH	CSGARCH	-1.1371	-1.3322	0.9379	10.3954	-9.3275
APARCH	TGARCH	-0.5398	1.2468	0.2496	10.4013	-592.3832
APARCH	AVGARCH	-1.7974	-2.5886	-1.2039	10.5128	-331.4456
APARCH	NGARCH	-1.7128	2.1703	-0.3763	10.4806	-85.3791
APARCH	NAGARCH	0.0414	1.0768	1.8855	10.6379	-238.6173
APARCH	ALLGARCH	-0.5795	0.2841	3.8475	10.4214	-242.9756
IGARCH	CSGARCH	-2.8572	2.3414	-0.6351	-4.4668	8.7701
IGARCH	TGARCH	-0.5743	2.1490	-1.2354	-3.4557	-3.7371
IGARCH	AVGARCH	-2.8716	-0.7041	-1.6224	-4.3503	-28.0902
IGARCH	NGARCH	-4.9401	3.3385	-1.3231	-4.3859	16.8367
IGARCH	NAGARCH	0.2734	2.4374	-0.0827	-5.0218	-59.5047
IGARCH	ALLGARCH	-0.2130	2.2610	3.0984	-3.4774	-61.6759
CSGARCH	TGARCH	0.4769	1.7096	-0.9530	0.0675	-9.0154
CSGARCH	AVGARCH	-0.1828	-2.8755	-1.2805	-0.9838	-14.2221
CSGARCH	NGARCH	-0.7206	3.1317	-0.9964	-1.0121	-3.0992
CSGARCH	NAGARCH	1.8937	2.0153	0.4342	0.7285	-24.3817
CSGARCH	ALLGARCH	0.7736	1.8086	3.3350	-0.0061	-24.9870
TGARCH	AVGARCH	-0.7632	-3.0672	-1.1970	-1.4410	-67.1662
TGARCH	NGARCH	-0.8934	2.4486	-0.3985	-1.4307	34.7782
TGARCH	NAGARCH	0.4923	0.6056	1.9654	0.5134	-105.5284
TGARCH	ALLGARCH	0.3138	-0.3297	3.8700	-0.9839	-109.5189
AVGARCH	NGARCH	-0.6479	4.6448	1.7908	0.5031	142.4204
AVGARCH	NAGARCH	-0.6479	3.3905	2.2365	1.1601	-143.4768
AVGARCH	ALLGARCH	1.4732	3.1409	3.7540	1.3730	-151.6081
NGARCH	NAGARCH	1.8320	-1.5833	1.9860	1.1584	-1.0877e+07
NGARCH	ALLGARCH	1.3519	-5.4222	3.6861	1.3451	-1.1187e+07
NAGARCH	ALLGARCH	-0.3166	-0.9392	3.5154	-0.5580	-2.0636e+15

Table 14: Diebold-Mariano test: model comparison for Ripple

Model 1	Model 2	Normal	Skew. Normal	Student t	Skew. Student t	SU
SGARCH	EGARCH	0.4589	-1.8021	0.9480	1.1042	-338.0355
SGARCH	GJRGARCH	1.0201	-4.1827	-0.0778	2.2131	-90.5425
SGARCH	APARCH	0.1863	0.8992	-0.3127	-10.3759	247.9776
SGARCH	IGARCH	2.7194	-3.4802	-0.8112	5.6830	94.4972
SGARCH	CSGARCH	0.3687	-2.2275	-0.5993	0.1157	-129.3268
SGARCH	TGARCH	4.1934	0.7577	-0.3590	-1.0654	-403.8368
SGARCH	AVGARCH	1.6358	-3.1736	-0.0191	0.2103	-380.4686
SGARCH	NGARCH	1.6098	3.8039	0.0137	0.1527	289.8863
SGARCH	NAGARCH	-0.2442	-5.3768	-0.4062	-1.7888	-1.7561e+03
SGARCH	ALLGARCH	-0.3965	3.9812	0.9344	-0.7952	208.3464
EGARCH	GJRGARCH	0.1879	3.9812	-1.1811	-0.3802	327.5210
EGARCH	APARCH	-0.2476	1.5956	-1.0539	-11.0590	350.4703
EGARCH	IGARCH	-0.3808	-2.1322	-1.2307	1.5902	345.1496
EGARCH	CSGARCH	-0.3828	-1.7138	-1.1578	-1.0609	107.3656
EGARCH	TGARCH	3.5486	2.0197	-1.0545	-1.3951	155.6681
EGARCH	AVGARCH	0.7938	-1.7611	-0.9632	-0.9855	168.3958
EGARCH	NGARCH	0.5670	3.8061	-0.9548	-1.0112	352.6339
EGARCH	NAGARCH	-0.6065	-0.4112	-0.9937	-1.4922	252.7224
EGARCH	ALLGARCH	-0.7404	2.5501	-0.1808	-1.3201	348.6361
GJRGARCH	APARCH	-1.1270	3.9161	-0.1372	-10.7642	150.1535
GJRGARCH	IGARCH	-0.9345	4.8030	-0.5171	4.9121	363.7847
GJRGARCH	CSGARCH	-0.9138	2.1934	-0.3440	-1.6555	-122.9948
GJRGARCH	TGARCH	3.6180	4.3551	-0.1436	-2.5812	-383.5916
GJRGARCH	AVGARCH	1.0893	3.8028	0.0553	-1.3069	-358.9501
GJRGARCH	NGARCH	0.7290	5.7420	0.0730	-1.3870	165.5345
GJRGARCH	NAGARCH	-1.3669	2.3364	-0.0149	-3.2587	-574.2380
GJRGARCH	ALLGARCH	-1.7494	4.7288	1.2523	-2.3786	136.0195
APARCH	IGARCH	-0.1042	-3.1831	-0.2637	11.4953	-21.1096
APARCH	CSGARCH	-0.1121	-1.8967	-0.1392	10.1578	-137.0913
APARCH	TGARCH	3.6859	-0.7979	-0.0143	10.1601	-408.6651
APARCH	AVGARCH	2.0077	-2.5365	0.9263	10.2430	-386.9312
APARCH	NGARCH	1.6420	0.6773	1.0714	10.2161	3.3756e+03
APARCH	NAGARCH	-0.4578	-2.9529	0.1983	10.0951	-1.7538e+05
APARCH	ALLGARCH	-0.7183	-0.1393	1.0331	10.2250	-3.2873e+03
IGARCH	CSGARCH	-0.0515	0.6128	0.5924	-5.5531	-135.3507
IGARCH	TGARCH	4.1693	3.7019	0.2807	-6.3866	-403.3644
IGARCH	AVGARCH	1.5956	1.7966	0.5129	-5.8674	-380.4508
IGARCH	NGARCH	1.5545	5.3946	0.5171	-5.9490	37.4971
IGARCH	NAGARCH	-0.3319	1.3128	0.6614	-5.7005	-723.6208
IGARCH	ALLGARCH	-0.4911	4.1473	1.2476	-6.2181	7.0627
CSGARCH	TGARCH	4.1727	2.3249	0.1475	-1.0903	-5.0695
CSGARCH	AVGARCH	1.5829	0.5854	0.3791	0.1624	9.5226
CSGARCH	NGARCH	1.5266	3.4525	0.3877	0.0981	138.7994
CSGARCH	NAGARCH	-0.3126	0.6494	0.4461	-1.4826	59.8795
CSGARCH	ALLGARCH	-0.4655	2.7321	1.1651	-0.8156	135.6411
TGARCH	AVGARCH	-3.7022	-3.5626	1.0659	2.0627	1.2328e+03
TGARCH	NGARCH	-3.8529	5.2495	1.1219	1.9611	413.5712
TGARCH	NAGARCH	-3.6314	-4.5842	0.2284	-0.1696	184.4384
TGARCH	ALLGARCH	-3.6811	1.5343	1.0370	1.5937	404.4039
AVGARCH	NGARCH	-1.6373	6.2072	0.4334	-0.7229	391.9593
AVGARCH	NAGARCH	1.3509	0.4406	-0.1117	-1.0883	157.4188
AVGARCH	ALLGARCH	-1.5512	4.0286	0.9271	-1.6652	382.5751
NGARCH	NAGARCH	-1.1906	-5.4224	-0.1343	-1.0877	-1.1793e+06
NGARCH	ALLGARCH	-1.4132	-1.8839	0.9166	-1.5249	-4.7192e+04
NAGARCH	ALLGARCH	-0.5842	5.6198	0.9946	0.3478	5.1937e+15

Table 15: Diebold-Mariano test: distribution comparison for Bitcoin, Litecoin and Ripple (1/2)

Model	Distribution 1	Distribution 2	BTC	LTC	RIPP
SGARCH	Normal	Skew. Normal	-5.5831	-7.7502	-6.0814
SGARCH	Normal	Student t	-3.3109	-1.7839	-5.0532
SGARCH	Normal	Skew. Student t	-7.9300	-6.8385	-10.3226
SGARCH	Normal	SU	-38.9903	-94.5863	-75.2867
SGARCH	Skew. Normal	Student t	5.5674	7.4649	5.8868
SGARCH	Skew. Normal	Skew. Student t	5.4362	6.8825	5.5658
SGARCH	Skew. Normal	SU	4.3064	-0.7434	1.2041
SGARCH	Student t	Skew. Student t	-8.0773	-7.6292	-7.6192
SGARCH	Student t	SU	-51.0946	-135.0892	-58.6960
SGARCH	Skew. Student t	SU	-37.5151	-92.2301	-45.8826
EGARCH	Normal	Skew. Normal	-5.1846	-5.8859	-5.4579
EGARCH	Normal	Student t	-2.3260	-2.3089	-4.1813
EGARCH	Normal	Skew. Student t	-5.9955	-7.8572	-9.6631
EGARCH	Normal	SU	-85.6213	-166.8948	-250.5264
EGARCH	Skew. Normal	Student t	5.0884	5.7898	5.3455
EGARCH	Skew. Normal	Skew. Student t	4.8025	5.5265	5.0765
EGARCH	Skew. Normal	SU	-2.3978	-1.6538	-4.0237
EGARCH	Student t	Skew. Student t	-9.0303	-7.7699	-7.4408
EGARCH	Student t	SU	-163.0394	-190.5718	-188.6318
EGARCH	Skew. Student t	SU	-128.3375	-149.5350	-163.4369
GJRGARCH	Normal	Skew. Normal	-6.5245	-8.3573	-6.8818
GJRGARCH	Normal	Student t	-3.5299	-2.0635	-5.6125
GJRGARCH	Normal	Skew. Student t	-7.7812	-6.6592	-10.3690
GJRGARCH	Normal	SU	-59.2478	-140.7995	-79.3782
GJRGARCH	Skew. Normal	Student t	6.4025	8.1076	6.7875
GJRGARCH	Skew. Normal	Skew. Student t	6.1300	7.6403	6.5878
GJRGARCH	Skew. Normal	SU	3.5024	-0.3584	5.0467
GJRGARCH	Student t	Skew. Student t	-7.4336	-6.2435	-7.1516
GJRGARCH	Student t	SU	-79.4559	-226.6460	-66.4141
GJRGARCH	Skew. Student t	SU	-58.3000	-127.7611	-49.0332
APARCH	Normal	Skew. Normal	-4.8220	-5.1229	-2.0366
APARCH	Normal	Student t	-2.5818	-2.9343	-5.3192
APARCH	Normal	Skew. Student t	-10.0118	-17.4986	-23.7390
APARCH	Normal	SU	-35.6781	-78.2404	-67.4486
APARCH	Skew. Normal	Student t	4.8280	5.1314	1.9300
APARCH	Skew. Normal	Skew. Student t	4.2868	3.8119	1.2691
APARCH	Skew. Normal	SU	2.4083	-1.1595	-0.1324
APARCH	Student t	Skew. Student t	-17.7009	-19.0634	-17.4820
APARCH	Student t	SU	-69.6883	-92.3492	-53.8175
APARCH	Skew. Student t	SU	-135.9648	-168.8279	-102.1709
IGARCH	Normal	Skew. Normal	-5.8329	-6.5650	-6.8270
IGARCH	Normal	Student t	-3.2433	-2.1656	-5.4910
IGARCH	Normal	Skew. Student t	-7.0827	-5.7263	-8.0720
IGARCH	Normal	SU	-28.3674	-104.8496	-69.5464
IGARCH	Skew. Normal	Student t	5.6901	6.3282	6.7001
IGARCH	Skew. Normal	Skew. Student t	5.2973	5.9965	6.5015
IGARCH	Skew. Normal	SU	3.5013	0.2062	4.7539
IGARCH	Student t	Skew. Student t	-7.5130	-5.9562	-5.3873
IGARCH	Student t	SU	-37.8734	-145.2296	-53.7599
IGARCH	Skew. Student t	SU	-25.8655	-99.5224	-42.2614
CSGARCH	Normal	Skew. Normal	-6.0017	-7.3136	-4.4359
CSGARCH	Normal	Student t	-2.6706	-1.9342	-5.4646
CSGARCH	Normal	Skew. Student t	-4.9395	-7.2969	-10.2344
CSGARCH	Normal	SU	-69.9978	-57.6859	-223.9766
CSGARCH	Skew. Normal	Student t	5.8728	7.0276	4.3675
CSGARCH	Skew. Normal	Skew. Student t	5.6093	6.4880	4.2065
CSGARCH	Skew. Normal	SU	-3.1062	2.8812e-04	0.8451
CSGARCH	Student t	Skew. Student t	-0.7412	-7.9913	-7.6027
CSGARCH	Student t	SU	-171.5110	-64.1463	-214.2942
CSGARCH	Skew. Student t	SU	-89.4242	-56.4201	-134.4837

Table 16: Diebold-Mariano test: distribution comparison for Bitcoin, Litecoin and Ripple (2/2)

Model	Distribution 1	Distribution 2	BTC	LTC	RIPP
TGARCH	Normal	Skew. Normal	-6.4305	-6.3123	-8.6284
TGARCH	Normal	Student t	-5.6129	-2.1800	-6.6134
TGARCH	Normal	Skew. Student t	-10.2664	-7.0339	-10.0269
TGARCH	Normal	SU	-137.2080	-94.6548	-136.0981
TGARCH	Skew. Normal	Student t	6.2849	6.5170	8.1693
TGARCH	Skew. Normal	Skew. Student t	6.1434	6.5170	7.8092
TGARCH	Skew. Normal	SU	2.3800	-5.2307	-8.2080
TGARCH	Student t	Skew. Student t	-8.8959	-7.6080	-7.7130
TGARCH	Student t	SU	-140.1264	-126.0859	-162.9488
TGARCH	Skew. Student t	SU	-101.7136	-87.1896	-122.1970
AVGARCH	Normal	Skew. Normal	-6.7366	-8.9141	-8.5106
AVGARCH	Normal	Student t	-2.6319	-2.2500	-5.4901
AVGARCH	Normal	Skew. Student t	-6.4579	-7.9161	-10.6273
AVGARCH	Normal	SU	-69.4680	-105.2376	-183.6255
AVGARCH	Skew. Normal	Student t	6.6620	8.6724	8.3264
AVGARCH	Skew. Normal	Skew. Student t	6.3531	8.0452	7.8829
AVGARCH	Skew. Normal	SU	1.0848	0.1300	1.1290
AVGARCH	Student t	Skew. Student t	-9.0274	-7.8522	-7.5939
AVGARCH	Student t	SU	-141.0730	-130.7385	-154.3297
AVGARCH	Skew. Student t	SU	-105.9834	-94.0364	-115.8039
NGARCH	Normal	Skew. Normal	-2.2435	-8.0468	-17.9017
NGARCH	Normal	Student t	-3.0058	-2.0488	-5.4506
NGARCH	Normal	Skew. Student t	-5.9463	-7.7117	-10.5402
NGARCH	Normal	SU	-47.2120	-80.9312	-65.7299
NGARCH	Skew. Normal	Student t	2.1541	8.3061	15.9267
NGARCH	Skew. Normal	Skew. Student t	2.0463	7.4018	15.5917
NGARCH	Skew. Normal	SU	0.3798	-10.7007	-11.3529
NGARCH	Student t	Skew. Student t	-6.4055	-7.8452	-7.5878
NGARCH	Student t	SU	-72.2853	-103.5153	-52.6473
NGARCH	Skew. Student t	SU	-54.9298	-75.2947	-39.8719
NAGARCH	Normal	Skew. Normal	-6.3144	-5.7654	-6.7061
NAGARCH	Normal	Student t	-3.3121	-2.2716	-5.0590
NAGARCH	Normal	Skew. Student t	-7.8423	-7.3035	-10.5073
NAGARCH	Normal	SU	-39.2159	-99.7882	-121.7324
NAGARCH	Skew. Normal	Student t	6.2966	5.7948	6.5683
NAGARCH	Skew. Normal	Skew. Student t	6.1532	5.2935	6.3894
NAGARCH	Skew. Normal	SU	4.6755	-8.1989	1.8408
NAGARCH	Student t	Skew. Student t	-8.0729	-7.5469	-7.6929
NAGARCH	Student t	SU	-52.8015	-132.8448	-96.4200
NAGARCH	Skew. Student t	SU	-39.7743	-101.8999	-77.4407
ALLGARCH	Normal	Skew. Normal	-6.9484	-8.3520	-4.4191
ALLGARCH	Normal	Student t	-2.4270	-0.8185	-3.8480
ALLGARCH	Normal	Skew. Student t	-6.3445	-8.0759	-10.5147
ALLGARCH	Normal	SU	-39.2206	-118.4434	-69.2232
ALLGARCH	Skew. Normal	Student t	6.7997	8.4022	4.2373
ALLGARCH	Skew. Normal	Skew. Student t	6.5791	7.7135	3.8622
ALLGARCH	Skew. Normal	SU	3.2500	-9.9147	-0.0442
ALLGARCH	Student t	Skew. Student t	-8.5570	-8.4354	-7.2585
ALLGARCH	Student t	SU	-78.4962	-124.4531	-53.7805
ALLGARCH	Skew. Student t	SU	-55.0671	-102.1780	-41.7529

Table 17: Unconditional Coverage, Independence and Conditional Coverage test results for VaR of Bitcoin

Model	UC 0.05	CC 0.05	IND 0.05	UC 0.01	CC 0.01	IND 0.01
Norm. SGARCH	-1.1170 (1)	2.2177 (0.3299)	3.3347 (0.0678)	-0.1248 (1)	2.1227 (0.3460)	2.2475 (0.1338)
Norm. EGARCH	-1.9491 (1)	6.9003 (0.0317)	8.8495 (0.0029)	1.1538 (0.2828)	5.3207 (0.0699)	4.1669 (0.0412)
Norm. GJRARCH	-1.1170 (1)	2.2177 (0.3299)	3.3347 (0.0678)	-0.1248 (1)	2.1227 (0.3460)	2.2475 (0.1338)
Norm. APARCH	-1.7903 (1)	1.6087 (0.4474)	3.3990 (0.0652)	1.1538 (0.2828)	5.3207 (0.0699)	4.1669 (0.0412)
Norm. IGARCH	-0.5885 (1)	2.5812 (0.2751)	3.1697 (0.0750)	-1.1116 (1)	1.0691 (0.5859)	2.1807 (0.1397)
Norm. CSGARCH	-1.1170 (1)	1.0685 (0.5961)	2.1855 (0.1393)	-0.1248 (1)	2.1227 (0.3460)	2.2475 (0.1338)
Norm. TGARCH	4.3486 (0.0370)	6.8602 (0.0324)	2.5117 (0.0370)	-1.1116 (1)	1.0691 (0.5859)	2.1807 (0.1397)
Norm. AVGARCH	-1.7903 (1)	1.6087 (0.4474)	3.3990 (0.0652)	1.1538 (0.2828)	5.3207 (0.0699)	4.1669 (0.0412)
Norm. NGARCH	-1.1170 (1)	1.0685 (0.5961)	2.1855 (0.1393)	1.1538 (0.2828)	3.4786 (0.1756)	2.3249 (0.1273)
Norm. NAGARCH	-1.1170 (1)	2.2177 (0.3299)	3.3347 (0.0678)	-0.1248 (1)	2.1227 (0.3460)	2.2475 (0.1338)
Norm. ALLGARCH	-1.7903 (1)	1.6087 (0.4474)	3.3990 (0.0652)	1.1538 (0.2828)	5.3207 (0.0699)	4.1669 (0.0412)
Skew. Norm. SGARCH	5.8877 (0.0152)	8.3007 (0.0158)	2.4129 (0.1203)	-1.7584 (1)	0.3660 (0.8328)	2.1244 (0.1450)
Skew. Norm. EGARCH	332.4069 (0)	334.4883 (0)	2.0814 (0.1419)	420.6271 (0)	426.3979 (0)	5.7708 (0.0163)
Skew. Norm. GJRARCH	895.6491 (0)	897.8599 (0)	2.2107 (0.1371)	955.8920 (0)	958.3804 (0)	2.4884 (0.1147)
Skew. Norm. APARCH	56.8705 (0)	59.7286 (0)	2.8581 (0.0909)	138.4541 (0)	143.3693 (0)	4.9152 (0.0266)
Skew. Norm. IGARCH	418.5993 (0)	421.2501 (0)	2.6508 (0.1035)	525.3967 (0)	527.7842 (0)	2.3875 (0.1223)
Skew. Norm. CSGARCH	408.6788 (0)	410.6800 (0)	2.0012 (0.1572)	496.8731 (0)	498.8782 (0)	2.0050 (0.1568)
Skew. Norm. TGARCH	28.3167 (0)	30.3255 (0)	2.0088 (0.1564)	86.0172 (0)	88.3804 (0)	2.3632 (0.1242)
Skew. Norm. AVGARCH	474.6609 (0)	478.0188 (0)	3.3579 (0.0669)	720.3862 (0)	722.4348 (0)	2.0485 (0.1524)
Skew. Norm. NGARCH	89.0956 (0)	91.1074 (0)	2.0118 (0.1561)	202.5935 (0)	204.8170 (0)	2.2235 (0.1359)
Skew. Norm. NAGARCH	21.2817 (0)	21.2817 (0)	21.2817 (0.1529)	6.0227 (0.0141)	8.0202 (0.0181)	1.9975 (0.1576)
Skew. Norm. ALLGARCH	11.9921 (0)	14.1728 (0)	2.1807 (0.1397)	-1.1116 (1)	1.0691 (0.5859)	2.1807 (0.1397)
Stud. SGARCH	30.8480 (0)	32.8505 (0)	2.0025 (0.1570)	6.0227 (0.0141)	8.0202 (0.0181)	1.9975 (0.1576)
Stud. EGARCH	7.6547 (0.0057)	11.8217 (0.0027)	4.1669 (0.0412)	-1.7261 (1)	0.3169 (0.8535)	2.0430 (0.1529)
Stud. GJRARCH	30.8480 (0)	32.8505 (0)	2.0025 (0.1570)	6.0227 (0.0141)	8.0202 (0.0181)	1.9975 (0.1576)
Stud. APARCH	7.6547 (0.0057)	11.8217 (0.0027)	4.1669 (0.0412)	-1.7261 (1)	0.3169 (0.8535)	2.0430 (0.1529)
Stud. IGARCH	11.9921 (0)	14.1728 (0)	2.1807 (0.1397)	1.2375 (0.2260)	3.2400 (0.1979)	0.2660 (0.1570)
Stud. CSGARCH	5.8877 (0.0152)	9.6309 (0.0081)	3.7432 (0.0530)	-1.7261 (1)	0.3169 (0.8535)	2.0430 (0.1529)
Stud. TGARCH	11.9921 (0)	14.1728 (0)	2.1807 (0.1397)	-1.7261 (1)	0.3169 (0.8535)	2.0430 (0.1529)
Stud. AVGARCH	11.9921 (0)	14.1728 (0)	2.1807 (0.1397)	-1.7261 (1)	0.3169 (0.8535)	2.0430 (0.1529)
Stud. NGARCH	11.9921 (0)	14.1728 (0)	2.1807 (0.1397)	-1.7261 (1)	0.3169 (0.8535)	2.0430 (0.1529)
Stud. NAGARCH	30.8480 (0)	32.8505 (0)	2.0025 (0.1570)	6.0227 (0.0141)	8.0202 (0.0181)	1.9975 (0.1576)
Stud. ALLGARCH	30.8480 (0)	32.8505 (0)	2.0025 (0.1570)	6.0227 (0.0141)	8.0202 (0.0181)	1.9975 (0.1576)
Skew. Stud. SGARCH	156.8275 (0)	159.4995 (0)	2.6720 (0.1021)	77.2618 (0)	79.3422 (0)	2.0803 (0.1492)
Skew. Stud. EGARCH	261.8362 (0)	263.9433 (0)	2.1071 (0.1466)	180.5007 (0)	183.1816 (0)	2.6809 (0.1016)
Skew. Stud. GJRARCH	136.1271 (0)	138.5426 (0)	2.4155 (0.1201)	56.6001 (0)	58.6160 (0)	2.0158 (0.1557)
Skew. Stud. APARCH	266.0733 (0)	268.1265 (0)	2.0533 (0.1519)	202.5935 (0)	204.5944 (0)	2.0009 (0.1572)
Skew. Stud. IGARCH	-1.5287 (1)	0.5664 (0.7534)	2.0951 (0.1478)	-1.7261 (1)	0.3169 (0.8535)	2.0430 (0.1529)
Skew. Stud. CSGARCH	1.5263e+03 (0)	1.5286e+03 (0)	2.2710 (0.1318)	2.4432e+03 (0)	2.4454e+03 (0)	2.2094 (0.1372)
Skew. Stud. TGARCH	266.0733 (0)	268.1265 (0)	2.0533 (0.1519)	213.8880 (0)	215.9700 (0)	2.0820 (0.1490)
Skew. Stud. AVGARCH	261.8362 (0)	264.1586 (0)	2.3224 (0.1275)	185.9603 (0)	188.2858 (0)	2.3255 (0.1273)
Skew. Stud. NGARCH	25.5391 (0)	27.5567 (0)	2.0176 (0.1555)	6.0227 (0.0141)	8.0202 (0.0181)	1.9975 (0.1576)
Skew. Stud. NAGARCH	139.5056 (0)	141.5366 (0)	2.0310 (0.1541)	68.7784 (0)	71.0402 (0)	2.2618 (0.1326)
Skew. Stud. ALLGARCH	300.8112 (0)	303.2908 (0)	2.4797 (0.1153)	236.9502 (0)	239.0778 (0)	2.1275 (0.1447)
SU. SGARCH	0.0786 (0.7792)	2.5198 (0.2837)	2.4412 (0.1182)	-1.7584 (1)	0.3660 (0.8328)	2.1244 (0.1450)
SU. EGARCH	605.5950 (0)	609.8467 (0)	4.2517 (0.0392)	1.1999e+03 (0)	1.2042e+03 (0)	4.2517 (0.0392)
SU. GJRARCH	7.6020 (0.0182)	2.0286 (0.0223)	0.0182 (0.1544)	2.6881 (0.1011)	5.1010 (0.0780)	2.4129 (0.1203)
SU. APARCH	2.3886e+03 (0)	2.3906e+03 (0)	1.9975 (0.1576)	3.3608e+03 (0)	3.3628e+03 (0)	2.0422 (0.1530)
SU. IGARCH	7.6547 (0.0057)	9.9796 (0.0068)	2.3249 (0.1273)	6.0227 (0.0141)	8.0202 (0.0181)	1.9975 (0.1576)
SU. CSGARCH	1.5090e+03 (0)	1.5110e+03 (0)	2.0270 (0.1545)	1.5079e+03 (0)	1.5099e+03 (0)	2.0141 (0.1558)
SU. TGARCH	2.3886e+03 (0)	2.3906e+03 (0)	1.9975 (0.1576)	3.2128e+03 (0)	3.2154e+03 (0)	2.5528 (0.1101)
SU. AVGARCH	2.3886e+03 (0)	2.3906e+03 (0)	1.9975 (0.1576)	3.6729e+03 (0)	3.6749e+03 (0)	1.9975 (0.1576)
SU. NGARCH	1.1641e+03 (0)	1.1672e+03 (0)	3.1107 (0.0778)	2.2231e+03 (0)	2.2257e+03 (0)	2.6213 (0.1054)
SU. NAGARCH	38.9346 (0)	40.9320 (0)	1.9975 (0.1576)	6.0227 (0.0141)	8.0202 (0.0181)	1.9975 (0.1576)
SU. ALLGARCH	2.3886e+03 (0)	2.3906e+03 (0)	1.9975 (0.1576)	6.0227 (0.0141)	8.0202 (0.0181)	1.9975 (0.1576)

Table 18: Unconditional Coverage, Independence and Conditional Coverage test results for VaR of Litecoin

Model	UC 0.05	CC 0.05	IND 0.05	UC 0.01	CC 0.01	IND 0.01
Norm. SGARCH	-1.5151 (1)	2.2036 (0.3323)	3.7187 (0.0538)	-1.1116 (1)	1.0691 (0.5859)	2.1807 (0.1397)
Norm. EGARCH	-1.7903 (1)	1.6087 (0.4474)	1.6087 (0.0652)	-1.1116 (1)	4.1750 (0.1240)	5.2867 (0.0215)
Norm. GJRARCH	-1.1170 (1)	1.0685 (0.5861)	2.1855 (0.1393)	-1.1116 (1)	1.0691 (0.5859)	2.1807 (0.1397)
Norm. APARCH	-1.9491 (1)	1.1695 (0.5573)	3.1186 (0.0774)	-1.1116 (1)	4.1750 (0.1240)	5.2867 (0.0215)
Norm. IGARCH	-0.5885 (1)	1.7097 (0.4254)	2.2982 (0.1295)	-1.9975(1)	0.0810 (0.9603)	2.0785 (0.1494)
Norm. CSGARCH	-1.7903 (1)	1.6087 (0.4474)	1.6087 (0.0652)	-1.9975(1)	0.0810 (0.9603)	2.0785 (0.1494)
Norm. TGARCH	1.8683 (0.1717)	4.6985 (0.0954)	2.8302 (0.0925)	-0.1248 (1)	2.1227 (0.3460)	2.2475 (0.1338)
Norm. AVGARCH	-1.1170 (1)	2.9642 (0.2272)	4.0811 (0.0434)	-1.7584 (1)	0.3360 (0.8328)	2.1244 (0.1450)
Norm. NGARCH	-1.5151 (1)	2.2036 (0.3323)	3.7187 (0.0538)	-1.7584 (1)	0.3360 (0.8328)	2.1244 (0.1450)
Norm. NAGARCH	-1.9491 (1)	3.5673 (0.1680)	3.5673 (0.0188)	1.1538 (0.2828)	3.4786 (0.1756)	2.3249 (0.1273)
Norm. ALLGARCH	-1.5151 (1)	0.5857 (0.7462)	2.1008 (0.1472)	-1.7584 (1)	0.3360 (0.8328)	2.1244 (0.1450)
Skew. Norm. SGARCH	566.2002 (0)	572.2192 (0)	6.0190 (0.0142)	767.6130 (0)	771.4310 (0)	3.8181 (0.0507)
Skew. Norm. EGARCH	433.6382 (0)	435.9248 (0)	2.2866 (0.1305)	539.8124 (0)	545.3036 (0)	5.4913 (0.0191)
Skew. Norm. GJRARCH	566.2002 (0)	576.1273 (0)	9.9271 (0.0016)	889.1263 (0)	900.5886 (0)	11.4623 (0)
Skew. Norm. APARCH	38.9346 (0)	40.9320 (0)	1.9975 (0.1576)	6.0227 (0.0141)	8.0202 (0.0181)	1.9975 (0.1576)
Skew. Norm. IGARCH	577.3552 (0)	582.4809 (0)	5.1257 (0.0236)	783.5327 (0)	786.1144 (0)	2.5817 (0.1081)
Skew. Norm. CSGARCH	549.6193 (0)	554.3738 (0)	4.7545 (0.0292)	743.8992 (0)	746.7061 (0)	2.8069 (0.0939)
Skew. Norm. TGARCH	1.8162 (0.1778)	3.8202 (0.1481)	2.0040 (0.1569)	24.7223 (0)	26.7645 (0)	2.0422 (0.1530)
Skew. Norm. AVGARCH	571.7676 (0)	575.9877 (0)	4.2201 (0.0399)	872.6454 (0)	877.2217 (0)	4.5763 (0.0324)
Skew. Norm. NGARCH	-1.1170 (1)	1.0685 (0.5861)	2.1855 (0.1393)	-1.1116 (1)	1.0691 (0.5859)	2.1807 (0.1397)
Skew. Norm. NAGARCH	30.8480 (0)	32.8505 (0)	2.0025 (0.1570)	6.0227 (0.141)	8.0202 (0.0181)	1.9975 (0.1576)
Skew. Norm. ALLGARCH	0.3753 (0.5401)	2.3911 (0.3025)	2.0158 (0.1557)	16.0200 (0)	19.1897 (0)	3.1697 (0.0750)
Stud. SGARCH	38.9346 (0)	40.9320 (0)	1.9975 (0.1576)	6.0227 (0.0141)	8.0202 (0.0181)	1.9975 (0.1576)
Stud. EGARCH	30.8480 (0)	32.8505 (0)	2.0025 (0.1570)	6.0227 (0.0141)	8.0202 (0.1811)	1.9975 (0.1576)
Stud. GJRARCH	38.9346 (0)	40.9320 (0)	1.9975 (0.1576)	6.0227 (0.0141)	8.0202 (0.1811)	1.9975 (0.1576)
Stud. APARCH	38.9346 (0)	40.9320 (0)	1.9975 (0.1576)	6.0227 (0.0141)	8.0202 (0.1811)	1.9975 (0.1576)
Stud. IGARCH	7.6547 (0.0057)	11.8217 (0.0027)	4.1669 (0.0412)	6.0227 (0.0141)	8.0202 (0.1811)	1.9975 (0.1576)
Stud. CSGARCH	5.8877 (0.0152)	9.6309 (0.0081)	3.7432 (0.0530)	1.2375 (0.2660)	3.2400 (0.1979)	2.0025 (0.1570)
Stud. TGARCH	38.9346 (0)	40.9320 (0)	1.9975 (0.1576)	6.0227 (0.0141)	8.0202 (0.1811)	1.9975 (0.1576)
Stud. AVGARCH	38.9346 (0)	40.9320 (0)	1.9975 (0.1576)	6.0227 (0.0141)	8.0202 (0.1811)	1.9975 (0.1576)
Stud. NGARCH	38.9346 (0)	40.9320 (0)	1.9975 (0.1576)	6.0227 (0.0141)	8.0202 (0.1811)	1.9975 (0.1576)
Stud. NAGARCH	38.9346 (0)	40.9320 (0)	1.9975 (0.1576)	6.0227 (0.0141)	8.0202 (0.1811)	1.9975 (0.1576)
Stud. ALLGARCH	38.9346 (0)	40.9320 (0)	1.9975 (0.1576)	6.0227 (0.0141)	8.0202 (0.1811)	1.9975 (0.1576)
Skew. Stud. SGARCH	305.2571 (0)	309.6020 (0)	4.3449 (0.0371)	81.6065 (0)	83.6350 (0)	2.0286 (0.1544)
Skew. Stud. EGARCH	398.8431 (0)	401.8778 (0)	3.0347 (0.0815)	169.7128 (0)	175.2591 (0)	5.5463 (0.0185)
Skew. Stud. GJRARCH	220.7897 (0)	223.2151 (0)	2.4254 (0.1194)	68.7784 (0)	70.8109 (0)	2.0325 (0.1540)
Skew. Stud. APARCH	398.8431 (0)	402.3605 (0)	3.5174 (0.0607)	169.7128 (0)	173.9053 (0)	4.1925 (0.0406)
Skew. Stud. IGARCH	-1.1825 (1)	1.0348 (0.5961)	2.2173 (0.1365)	-0.7701 (1)	1.2476 (0.5359)	2.0176 (0.1555)
Skew. Stud. CSGARCH	389.0926 (0)	392.9073 (0)	3.8147 (0.0508)	164.3859 (0)	170.4338 (0)	6.0478 (0.139)
Skew. Stud. TGARCH	398.8431 (0)	402.3605 (0)	3.5174 (0.0607)	159.1048 (0)	164.1088 (0)	5.0040 (0.0253)
Skew. Stud. AVGARCH	403.7503 (0)	407.1343 (0)	3.3840 (0.0658)	164.3859 (0)	168.9670 (0)	4.5810 (0.0323)
Skew. Stud. NGARCH	403.7503 (0)	407.1343 (0)	3.3840 (0.0658)	164.3859 (0)	168.9670 (0)	4.5810 (0.0323)
Skew. Stud. NAGARCH	296.3881 (0)	299.4067 (0)	3.0187 (0.0823)	81.6065 (0)	85.4813 (0)	3.8748 (0.0490)
Skew. Stud. ALLGARCH	398.8431 (0)	402.3605 (0)	3.5174 (0.0607)	169.7128 (0)	173.9053 (0)	4.1925 (0.0406)
SU. SGARCH	1.3024e+03 (0)	1.3044e+03 (0)	1.9992 (0.1574)	1.0672e+03(0)	1.0709e+03 (0)	3.6521 (0.0560)
SU. EGARCH	681.3681 (0)	686.1203 (0)	4.7521 (0.0293)	1.3186e+03 (0)	1.3234e+03 (0)	4.7521 (0.0293)
SU. GJRARCH	1.2631e+03 (0)	1.2651e+03 (0)	1.9984 (0.1575)	1.5564e+03 (0)	1.5596e+03 (0)	3.1674 (0.0751)
SU. APARCH	1.0542e+03 (0)	1.0563e+03 (0)	2.0963 (0.1477)	213.8880 (0)	216.6604 (0)	2.7723 (0.0959)
SU. IGARCH	1.1638e+03 (0)	1.1662e+03 (0)	2.3514 (0.1252)	1.3372e+03 (0)	1.3427e+03 (0)	5.4958 (0.0191)
SU. CSGARCH	1.0757e+03 (0)	1.0783e+03 (0)	2.5601 (0.1096)	1.1463e+03 (0)	1.1500e+03 (0)	3.6588 (0.0558)
SU. TGARCH	38.4404 (0)	41.9562 (0)	3.5158 (0.0608)	-0.1248 (1)	2.1227 (0.3460)	2.2475 (0.1338)
SU. AVGARCH	2.2631e+03 (0)	2.2654e+03 (0)	2.3249 (0.1273)	1.9006e+03 (0)	1.9030e+03 (0)	2.4556 (0.1171)
SU. NGARCH	2.3886e+03 (0)	2.3906e+03 (0)	1.9975 (0.1576)	3.6729e+03 (0)	3.6749e+03 (0)	1.9975 (0.1576)
SU. NAGARCH	2.3886e+03 (0)	2.3906e+03 (0)	1.9975 (0.1576)	3.6729e+03 (0)	3.6749e+03 (0)	1.9975 (0.1576)
SU. ALLGARCH	2.3886e+03 (0)	2.3906e+03 (0)	1.9975 (0.1576)	3.6729e+03 (0)	3.6749e+03 (0)	1.9975 (0.1576)

Table 19: Unconditional Coverage, Independence and Conditional Coverage test results for VaR of Ripple

Model	UC 0.05	CC 0.05	IND 0.05	UC 0.01	CC 0.01	IND 0.01
Norm. SGARCH	1.8683 (0.1717)	4.6102 (0.0998)	2.7418 (0.0978)	-1.7584 (1)	0.3660 (0.8328)	2.1244 (0.1450)
Norm. EGARCH	5.8877 (0.0152)	9.6309 (0.0081)	3.7432 (0.0530)	-1.7261 (1)	0.3169 (0.8535)	2.0430 (0.1529)
Norm. GJRARCH	1.8683 (0.1717)	4.6102 (0.0998)	2.7418 (0.0978)	-1.9975 (1)	0.0810 (0.9603)	2.0785 (0.1494)
Norm. APARCH	0.8939 (0.3444)	3.5112 (0.1728)	2.6173 (0.1057)	-1.9975 (1)	0.0810 (0.9603)	2.0785 (0.1494)
Norm. IGARCH	1.8683 (0.1717)	4.6102 (0.0998)	2.7418 (0.0978)	-1.7584 (1)	0.3660 (0.8328)	2.1244 (0.1450)
Norm. CSGARCH	0.8939 (0.3444)	3.7672 (0.1520)	2.8733 (0.0901)	-1.7584 (1)	0.3660 (0.8328)	2.1244 (0.1450)
Norm. TGARCH	-1.9491 (1)	1.9491 (0.3774)	3.8983 (0.0483)	-1.7584 (1)	0.3660 (0.8328)	2.1244 (0.1450)
Norm. AVGARCH	3.0148 (0.0825)	5.6361 (0.0825)	2.6213 (0.1054)	-1.7261 (1)	0.3169 (0.8535)	2.0430 (0.1529)
Norm. NGARCH	3.0148 (0.0825)	5.6361 (0.0825)	2.6213 (0.1054)	-1.9975 (1)	0.0810 (0.9603)	2.0785 (0.1494)
Norm. NAGARCH	-0.5885 (1)	1.7097 (0.4254)	2.2982 (0.1295)	-1.9975 (1)	0.0810 (0.9603)	2.0785 (0.1494)
Norm. ALLGARCH	-0.5885 (1)	1.7097 (0.4254)	2.2982 (0.1295)	-1.9975 (1)	0.0810 (0.9603)	2.0785 (0.1494)
Skew. Norm. SGARCH	38.9346 (0)	40.9320 (0)	1.9975 (0.1576)	6.0227 (0.0141)	8.0202 (0.0181)	1.9975 (0.1576)
Skew. Norm. EGARCH	522.3895 (0)	530.1281 (0)	7.7386 (0.0054)	673.9717 (0)	676.2074 (0)	2.2357 (0.1349)
Skew. Norm. GJRARCH	675.4191 (0)	686.8017 (0)	11.3826 (0)	989.7751 (0)	996.0993 (0)	6.3242 (0.0119)
Skew. Norm. APARCH	42.7997 (0)	45.2656 (0)	2.4658 (0.1163)	77.2618 (0)	84.8468 (0)	7.5850 (0.0059)
Skew. Norm. IGARCH	663.5812 (0)	674.9421 (0)	11.3609 (0)	856.2495 (0)	860.1204 (0)	3.8708 (0.0491)
Skew. Norm. CSGARCH	617.0317 (0)	625.2212 (0)	8.1895 (0.0042)	759.6862 (0)	762.7880 (0)	3.1017 (0.0782)
Skew. Norm. TGARCH	7.8564 (0.0051)	10.3543 (0.0056)	2.4979 (0.1140)	34.4764 (0)	36.4854 (0)	2.0090 (0.1564)
Skew. Norm. AVGARCH	634.3373 (0)	643.4581 (0)	9.1208 (0.0025)	856.2495 (0)	860.5124 (0)	4.2629 (0.0390)
Skew. Norm. NGARCH	-1.5151 (1)	2.2036 (0.3323)	3.7187 (0.0538)	1.1538 (0.2828)	3.4786 (0.1756)	2.3249 (0.1273)
Skew. Norm. NAGARCH	7.6547 (0.0057)	9.9796 (0.0068)	2.3249 (0.1273)	-0.7701 (1)	1.2476 (0.5359)	2.0176 (0.1555)
Skew. Norm. ALLGARCH	38.9346 (0)	40.9320 (0)	1.9975 (0.1576)	6.0227 (0.0141)	8.0202 (0.0181)	1.9975 (0.1576)
Stud. SGARCH	21.2817 (0)	23.3247 (0)	2.0430 (0.1529)	6.0227 (0.0141)	8.0202 (0.0181)	1.9975 (0.1576)
Stud. EGARCH	38.9346 (0)	40.9320 (0)	1.9975 (0.1576)	6.0227 (0.0141)	8.0202 (0.0181)	1.9975 (0.1576)
Stud. GJRARCH	38.9346 (0)	40.9320 (0)	1.9975 (0.1576)	6.0227 (0.0141)	8.0202 (0.0181)	1.9975 (0.1576)
Stud. APARCH	25.5391 (0)	27.5567 (0)	2.0176 (0.1555)	6.0227 (0.0141)	8.0202 (0.0181)	1.9975 (0.1576)
Stud. IGARCH	11.9921 (0)	14.1728 (0)	2.1807 (0.1397)	6.0227 (0.0141)	8.0202 (0.0181)	1.9975 (0.1576)
Stud. CSGARCH	11.9921 (0)	14.1728 (0)	2.1807 (0.1397)	6.0227 (0.0141)	8.0202 (0.0181)	1.9975 (0.1576)
Stud. TGARCH	25.5391 (0)	27.5567 (0)	2.0176 (0.1555)	6.0227 (0.0141)	8.0202 (0.0181)	1.9975 (0.1576)
Stud. AVGARCH	25.5391 (0)	27.5567 (0)	2.0176 (0.1555)	6.0227 (0.0141)	8.0202 (0.0181)	1.9975 (0.1576)
Stud. NGARCH	25.5391 (0)	27.5567 (0)	2.0176 (0.1555)	6.0227 (0.0141)	8.0202 (0.0181)	1.9975 (0.1576)
Stud. NAGARCH	21.2817 (0)	23.3247 (0)	2.0430 (0.1529)	6.0227 (0.0141)	8.0202 (0.0181)	1.9975 (0.1576)
Stud. ALLGARCH	21.2817 (0)	23.3247 (0)	2.0430 (0.1529)	6.0227 (0.0141)	8.0202 (0.0181)	1.9975 (0.1576)
Skew. Stud. SGARCH	479.8820 (0)	485.2464 (0)	5.3644 (0.0206)	278.7436 (0)	280.8023 (0)	2.0587 (0.1513)
Skew. Stud. EGARCH	418.5993 (0)	423.3431 (0)	4.7438 (0.0294)	231.1269 (0)	234.6652 (0)	3.5383 (0.0600)
Skew. Stud. GJRARCH	438.6932 (0)	442.4642 (0)	3.7710 (0.0521)	242.8113 (0)	244.8822 (0)	2.0709 (0.1501)
Skew. Stud. APARCH	253.4334 (0)	255.7094 (0)	2.2760 (0.1314)	95.0299 (0)	97.6826 (0)	2.6527 (0.1034)
Skew. Stud. IGARCH	86.2164 (0)	88.3286 (0)	2.1122 (0.1461)	10.8997 (0)	13.7730 (0.0010)	2.8733 (0.0901)
Skew. Stud. CSGARCH	490.3860 (0)	496.2629 (0)	5.8770 (0.0153)	278.7436 (0)	280.8023 (0)	2.0587 (0.1513)
Skew. Stud. TGARCH	490.3860 (0)	496.2629 (0)	5.8770 (0.0153)	284.8559 (0)	287.0674 (0)	2.2115 (0.1370)
Skew. Stud. AVGARCH	490.3860 (0)	496.2629 (0)	5.8770 (0.0153)	303.3972 (0)	305.4529 (0)	2.0557 (0.1516)
Skew. Stud. NGARCH	490.3860 (0)	496.2629 (0)	5.8770 (0.0153)	303.3972 (0)	305.4529 (0)	2.0557 (0.1516)
Skew. Stud. NAGARCH	464.2807 (0)	468.6363 (0)	4.3556 (0.0369)	260.6159 (0)	262.9944 (0)	2.3784 (0.1230)
Skew. Stud. ALLGARCH	500.9720 (0)	507.4133 (0)	6.4413 (0.0111)	284.8559 (0)	287.0674 (0)	2.2115 (0.1370)
SU. SGARCH	-1.9491 (1)	1.1695 (0.5573)	3.1186 (0.0774)	6.0227 (0.0141)	8.0202 (0.0181)	1.9975 (0.1576)
SU. EGARCH	760.5332 (0)	769.3588 (0)	8.8256 (0.0030)	1.4407e+03 (0)	1.4495e+03 (0)	8.8256 (0.030)
SU. GJRARCH	-1.1170 (1)	2.9642 (0.2272)	4.0811 (0.0434)	6.0227 (0.0141)	8.0202 (0.0181)	1.9975 (0.1576)
SU. APARCH	38.9346 (0)	40.9320 (0)	1.9975 (0.1576)	6.0227 (0.0141)	8.0202 (0.0181)	1.9975 (0.1576)
SU. IGARCH	25.5391 (0)	27.5567 (0)	2.0176 (0.1555)	6.0227 (0.0141)	8.0202 (0.0181)	1.9975 (0.1576)
SU. CSGARCH	1.3748e+03 (0)	1.3770e+03 (0)	2.1846 (0.1394)	1.5957e+03 (0)	1.5991e+03 (0)	3.4735 (0.624)
SU. TGARCH	1.8425e+03 (0)	1.8478e+03 (0)	5.3065 (0.0212)	1.8692e+03 (0)	1.8730e+03 (0)	3.8274 (0.0504)
SU. AVGARCH	1.8325e+03 (0)	1.8373e+03 (0)	4.8608 (0.0275)	1.8484e+03 (0)	1.8525e+03 (0)	4.0562 (0.0440)
SU. NGARCH	2.3886e+03 (0)	2.3906e+03 (0)	1.9975 (0.1576)	3.6729e+03 (0)	3.6749e+03 (0)	1.9975 (0.1576)
SU. NAGARCH	2.3886e+03 (0)	2.3906e+03 (0)	1.9975 (0.1576)	6.0227 (0.0141)	8.0202 (0.0181)	1.9975 (0.1576)
SU. ALLGARCH	2.3886e+03 (0)	2.3906e+03 (0)	1.9975 (0.1576)	6.0227 (0.0141)	8.0202 (0.0181)	1.9975 (0.1576)

References

- Akaike, H. (1974). A new look at the statistical model identification. *IEEE Transactions on Automatic Control* 19, 716–23.
- Azzalini, A. (1985). A class of distributions which includes the normal ones. *Scandinavian Journal of Statistics* 12, 171–78.
- Bollerslev, T. (1986). Generalized autoregressive conditional heteroskedasticity. *Journal of Econometrics* 31, 307–27.
- Bozdogan, H. (1987). Model selection and Akaike’s information criterion (aic): The general theory and its analytical extensions. *Psychometrika* 52, 345–70.
- Christoffersen, P. F. (1998). Evaluating interval forecasts. *International Economic Review* 39, 841–62.
- Chu, J., S. Chan, S. Nadarajah, and J. Osterrieder (2017). GARCH modeling of cryptocurrencies. *Journal of Risk and Financial Management*, 841–62.
- Diebold, F. and R. Mariano (1995). Comparing predictive accuracy. *Journal of Business and Economic Statistics* 13, 253–63.
- Diks, C., V. Panchenko, and D. van Dijk (2011). Likelihood-based scoring rules for comparing density forecasts in tails. *Journal of Econometrics* 163, 215–30.
- Ding, Z., C. W. Granger, and R. F. Engle (1993). A long memory property of stock market returns and a new model. *Journal of Empirical Finance* 1, 83–106.
- Engle, R. F. and T. Bollerslev (1986). Modelling the persistence of conditional variances. *Econometric Reviews* 5, 1–50.
- Engle, R. F. and V. K. Ng (1993). Measuring and testing the impact of news on volatility. *Journal of Finance* 48, 1749–78.
- Fernandez, C. and M. F. Steel (1998). On bayesian modelling of fat tails and skewness. *Journal of the American Statistical Association* 93, 359–71.
- Glosten, L. R., R. Jagannathan, and D. E. Runkle (1993). On the relation between the expected value and the volatility of the nominal excess return on stocks. *Journal of Finance* 48, 1779–801.
- Gosset, W. (1908). The probable error of a mean. *Biometrika* 6, 1–25.
- Hannan, E. and B. G. Quinn (1979). The determination of the order of an autoregression. *Journal of the Royal Statistical Society B41*, 190–95.
- Hentschel, L. (1995). All in the family nesting symmetric and asymmetric GARCH models. *Journal of Financial Economics* 39, 71–104.
- Higgins, M. L. and A. K. Bera. (1992). A class of nonlinear ARCH models. *International Economic Review* 6, 137–58.
- Hurvich, C. M. and C.-L. Tsai (1989). Regression and time series model selection in small samples. *Biometrika* 76, 297–307.

- Jarque, C. M. and A. K. Bera (1980). Efficient tests for normality, homoscedasticity and serial independence of regression residuals. *International Economic Review* 4.7, 313–18.
- Johnson, N. L. (1949). Systems of frequency curves generated by methods of translation. *Biometrika* 36, 149–76.
- Lee, G. J. and R. F. Engle (1999). A permanent and transitory component model of stock return volatility. *Cointegration Causality and Forecasting A Festschrift in Honor of Clive W. J. Granger, Oxford: Oxford University Press* 6, 475–97.
- Nelson, D. B. (1991). Conditional heteroskedasticity in asset returns: A new approach. *Econometrica* 59, 347–70.
- Owen, D. B. (1956). Tables for computing bivariate normal probabilities. *Annals of Mathematical Statistics* 27, 1075–90.
- Schwarz, G. E. (1978). Estimating the dimension of a model. *Annals of Statistics* 6, 461–64.
- Taylor, S. J. (1986). Modelling financial time series. *New York: John Wiley and Sons* 44.
- Zakoian, J.-M. (1994). Threshold heteroskedastic models. *Journal of Economic Dynamics and Control* 18, 931–55.